Ideal Private Simultaneous Messages Schemes and Their Applications

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Abstract

Private Simultaneous Messages (PSM) is a minimal model for secure computation, where two parties, Alice and Bob, have private inputs x, y and a shared random string. Each of them sends a single message to an external party, Charlie, who can compute $f(x, y)$ for a public function f but learns nothing else. The problem of narrowing the gap between upper and lower bounds on the communication complexity of PSM has been widely studied, but the gap still remains exponential. In this work, we study the communication complexity of PSM from a different perspective and introduce a special class of PSM, referred to as *ideal PSM*, in which each party's message length attains the minimum, that is, their messages are taken from the same domain as inputs. We initiate a systematic study of ideal PSM with a complete characterization, several positive results, and applications. First, we provide a characterization of the class of functions that admit ideal PSM, based on permutation groups acting on the input domain. This characterization allows us to derive asymptotic upper bounds on the total number of such functions and a complete list for small domains. We also present several infinite families of functions of practical interest that admit ideal PSM. Interestingly, by simply restricting the input domains of these ideal PSM schemes, we can recover most of the existing PSM schemes that achieve the best known communication complexity in various computation models. As applications, we show that these ideal PSM schemes yield novel communication-efficient PSM schemes for functions with sparse or dense truth-tables and those with low-rank truth-tables. Furthermore, we obtain a PSM scheme for general functions that improves the constant factor in the dominant term of the best known communication complexity. An additional advantage is that our scheme simplifies the existing construction by avoiding the hierarchical design of internally invoking PSM schemes for smaller functions.

1 Introduction

Secure computation [51] is a fundamental cryptographic primitive which enables two parties, Alice and Bob, to evaluate a function f over their joint inputs without revealing information other than the output. This problem has been considered in several different models and settings (see, e.g., [12, 17, 20–22, 31]). In this work, we consider this problem in a minimal model for communication in the setting of information-theoretic security, referred to as *Private Simultaneous Messages (PSM)* model [29, 36]. In a (two-party) PSM scheme, Alice holds an input x, Bob holds an input y, and they both share common randomness. Each of them sends a message to an external party, Charlie, who can then compute $f(x, y)$ but learns nothing else.

The central research question is to determine the minimum communication complexity of PSM schemes for a given function f, where communication complexity is defined as the total bit-length of Alice's and Bob's messages. This question has been extensively studied for general functions as well as for functions with specific representations [2, 3, 6, 7, 9, 23, 25, 34, 49]. The currently best known construction for a general function $f : X \times X \to \{0, 1\}$ provides a PSM scheme with communication complexity $O(\sqrt{N})$, where N denotes the cardinality of X [7]. Whereas, the lower bound of $3 \log N - O(\log \log N)$ bits is derived for certain functions [2]. Despite all the progresses, however, there still remains an exponential gap between upper and lower bounds.

In this work, we depart from prior approaches aimed at narrowing the exponential gap and study the communication complexity of PSM from a different perspective. It is clear that a communication complexity of $2 \log N$ bits is trivially unavoidable in light of the input length[1]. Thus, if a PSM scheme has communication complexity equal to the input length of f, then it is necessarily communication-optimal. We will refer to such schemes as *ideal PSM schemes*. Due to the lower bounds by [2, 23, 49], not every function admits ideal PSM. Nevertheless, some special functions of interest admit ideal PSM. For example, the function that computes the sum $x + y$ over an abelian group is realized by a PSM scheme where Alice sends $x + r$ and Bob sends $y - r$ for a uniformly random element r. This scheme is ideal since messages are taken from the same domain as inputs. The main goal of this work is to give a complete characterization of functions that admit ideal PSM.

This research direction is inspired by a series of works on *ideal secret sharing schemes* [11, 15, 18, 26–28, 44, 50]. A secret sharing scheme is called ideal if the size of every share is equal to that of a secret, which is proven to be the optimal size [38]. While a complete characterization is not known, several classes of access structures are proven to admit ideal secret sharing schemes (see Section 1.2 for details). We are the first to conduct a systematic study of ideal PSM schemes, in analogy with ideal secret sharing schemes.

1.1 Our Results

In this work, we initiate a theory of ideal PSM schemes with a complete characterization, several positive results, and applications. As an interesting implication, our results provide a unified perspective on existing PSM constructions achieving the best known communication complexity, and moreover yield novel and more efficient constructions. First, we show a necessary and sufficient condition for functions to admit ideal PSM, using group-theoretic terminology. Specifically, we prove that such functions are essentially limited to a family of functions induced by certain permutation groups on the input domain. This characterization allows us to derive asymptotic upper bounds on the number of functions admitting ideal PSM and a complete list when the domain size N is small (e.g., up to $N \leq 23$ for boolean functions). We also provide several infinite families of functions of special interest that admit ideal PSM schemes, including the sum function mentioned above. As applications, we propose novel communication-efficient PSM schemes for functions with sparse/dense or low-rank truth-tables, improving upon the previous construction for functions with bounded tiling number [43]. Furthermore, we reduce the constant factor in the dominant term of the best known communication complexity for general functions [7], from $12\sqrt{N}$ to $8\sqrt{N}$. Below, we elaborate on our results.

Characterization. We say that a PSM scheme is ideal if the message space of each party i is the same as X_i. A function $f : X_0 \times X_1 \to Y$ is called an ideal PSM function if there exists an ideal PSM scheme for f. To formalize our characterization, we introduce the *invariant group* I_f of f, defined as the set of all pairs of permutations $\pi = (\pi_0, \pi_1)$, where each π_i acts on X_i, such that $f(\pi_0(x), \pi_1(y)) = f(x, y)$ for all $(x, y) \in X_0 \times X_1$. This group naturally acts on $X_0 \times X_1$. We show that f is an ideal PSM function if and only if I_f satisfies the following condition: for any pair of inputs $(x, y), (x', y') \in X_0 \times X_1$ such that $f(x, y) = f(x', y')$, there exists a permutation $\pi \in I_f$ such that $(x', y') = \pi(x, y)$. This characterization makes it easier to test if a given function admits ideal PSM, by finding an appropriate permutation that preserves the outputs of f.

A key technical idea in our characterization is that the class of ideal PSM functions is essentially restricted to a special subclass, that is, every ideal PSM function is equivalent to one of them[2]. Specifically, each function $f_{\mathcal{G}}$ in the subclass is parameterized by a tuple $\mathcal{G} = (G_0, G_1, \psi)$, where G_i is a subgroup of permutations on X_i and ψ is an isomorphism from G_0 to G_1, and $f_{\mathcal{G}}$ maps each input (x, y) to its *orbit* under the action of $\mathcal{G}$ on $X_0 \times X_1$. We show that any function f admitting ideal PSM is equivalent to $f_{\mathcal{G}}$ for

[1]Note that if a non-zero probability of failure is allowed in the reconstruction, this lower bound does not apply. Throughout this paper, we focus on PSM with perfect correctness.

[2]We say that two functions are *equivalent* if they are identical up to permutations of inputs and outputs.

some $\mathcal{G}$. This formalization is useful for the enumeration of ideal PSM functions since it reduces the problem to the enumeration of all such tuples $\mathcal{G}$.

Enumeration. We provide asymptotic upper bounds on the total number of connected functions $f : X_0 \times X_1 \to Y$ that admit ideal PSM. Here, we additionally assume that f is connected, meaning that there exists no partition of $X_0 \times X_1$ into rectangles each of which corresponds to disjoint values of f. In the boolean case $Y = \{0, 1\}$, this assumption excludes only a few trivial functions and does not affect the asymptotic results. Note that naively enumerating all tuples $\mathcal{G}$ cannot yield a non-trivial upper bound. This is because a permutation group of degree N contains at least $2^{\Omega(N^2)}$ subgroups [45], implying that such a naive enumeration cannot improve upon the trivial bound of 2^{N^2} given by the total number of functions. We leverage group-theoretic properties of ideal PSM functions to remove duplicate counts of $\mathcal{G}$'s that correspond to equivalent functions. As a result, we obtain a non-trivial upper bound of $2^{O(N \log^2 N)}$ on the number of ideal PSM functions, where $N = \max\{|X_0|, |X_1|\}$. Furthermore, if $|X_0|$ and $|X_1|$ are primes, we can significantly improve the upper bound to $2^{O(\log^3 N)}$. These bounds demonstrate that the fraction of ideal PSM functions is exponentially small compared to the set of all functions.

We provide a complete list of ideal PSM functions for small domains by enumerating all tuples $\mathcal{G}$ determining non-equivalent functions $f_{\mathcal{G}}$. We use an open software package called GAP to enumerate permutation groups[3]. For a general range Y, we obtain a list of such functions up to $N \leq 15$ except for the case of $|X_0| = |X_1| = 12$. For a boolean case $Y = \{0, 1\}$, we show that each equivalence class of ideal PSM functions corresponds to a connected component of a certain graph, leading to substantial reduction in computational cost. As a result, we obtain a larger list of functions up to $N \leq 23$ in the boolean case. The resulting lists are available in [1].

Infinite Families. We show that the following families of practically relevant functions admit ideal PSM schemes:

- *Group product.* This function computes the iterated product of a combined sequence of (possibly non-commutative) group elements held by two parties.
- *Index function.* This function takes a function $\phi : H \to G$ from a certain class and a pair $(x, r) \in H \times G$ as input, and outputs $\phi(x) - r$, where H and G are abelian groups. This family generalizes the original notion of the index function [29, 39], which outputs D_x on input a vector $D = (D_i)_{1 \leq i \leq N}$ and an index x. It also includes the inner product and multi-linear polynomials as special cases.
- *Private set intersection cardinality (PSIC).* This function takes two subsets A_0 and A_1 of a common set as input and outputs $|A_0 \cap A_1|$. This family particularly includes the equality function, which tests the equality of two elements.

In particular, we are the first to consider PSIC functions in the PSM setting and to show a communication-optimal construction for them.

Interestingly, by simply restricting the input domains of the above ideal PSM schemes, we can recover most of the existing PSM schemes that achieve the state-of-the-art communication complexity in various computation models. The best known schemes for branching programs [29] and arithmetic formulas [19, 34] are obtained by evaluating the group product over restricted inputs. The nearly optimal construction for polynomials [39] is also obtained by computing a generalized index function, where ϕ is taken to be a multi-linear polynomial. As explained below, the previous constructions [3, 7] can be even refined and simplified within our framework based on ideal PSM schemes.

Communication-efficient and Simplified Constructions. We demonstrate that ideal PSM schemes for the above function families induce more communication-efficient PSM schemes for several functions. First, we consider a function $f : X \times X \to \{0, 1\}$ such that the number of 1's in each row of its truth-table T_f (viewed as a matrix) is at most $d \ll N := |X|$. Prior to our work, the only general construction in [7] can apply, resulting in communication complexity of $O(\sqrt{N})$. In contrast, by embedding T_f to the truth-table of a PSIC function, we obtain a novel PSM scheme for f with communication complexity $O(d \log(N + d))$, implying

[3] https://www.gap-system.org/about/

a strict improvement when $d = o(\sqrt{N}/\log N)$. The same communication complexity can be attained when the columns are sparse as well as when the rows (or columns) are dense, that is, they contain at least $n - d$ ones for a small d.

Next, Narayanan et al. [43] showed that a function $f : X \times X \to Y$ can be realized by a PSM scheme with communication complexity $O(k_f \log |Y|)$, where k_f denotes the tiling number of f. Here, the tiling number refers to the smallest number of disjoint monochromatic rectangles covering the truth-table T_f. On the other hand, by embedding T_f to the truth-table of the inner product, we present a novel PSM scheme for f with communication complexity $O(r_f \log |Y|)$, where r_f denotes the rank of T_f as a matrix over some field of size at least $|Y|$. As is well known, it always holds that $k_f \geq r_f$ (e.g., [32]) and thus our scheme achieves a more refined communication complexity.

Finally, Beimel et al. [7] presented a PSM scheme for a general function $f : X \times X \to \{0,1\}$ with communication complexity $O(\sqrt{N})$, which is at least $12\sqrt{N}$, where $N = |X|$. In contrast, we adopt a different approach based on ideal PSM schemes to improve and simplify their construction. We show an ideal PSM function $G_f : X' \times X' \to \{0,1\}$ with $\log |X'| = 4\sqrt{N} + 1$ that contains f as a restriction. As a result, we obtain a PSM scheme for f with communication complexity $8\sqrt{N} + 2$ from the ideal PSM scheme for G_f, improving the constant factor in the dominant term of [7]. An additional benefit is that our construction is simpler and more direct: Alice and Bob can compute their messages simply by applying explicitly given permutations to their inputs. As a result, the message functions of our scheme can be expressed in closed form and avoid the hierarchical design of [3,7], which relies on nested invocations of PSM schemes for smaller functions.

1.2 Related Work

We provide a brief history of the problem of determining the optimal communication complexity of PSM schemes. Feige et al. [29] formalized the notion of PSM schemes and presented the feasibility result of two-party PSM for general functions. Ishai and Kushilevitz [36] extended this to the multi-party setting and showed a k-party PSM scheme for functions represented by branching programs. These constructions, along with several variants including PSM for arithmetic formulas, are summarized in [34]. Subsequently, Beimel et al. [7] successfully devised a two-party PSM scheme for general functions $f : X \times X \to \{0,1\}$ with communication complexity $O(\sqrt{|X|})$. This result was later generalized and improved by [3,9], leading to k-party PSM schemes with communication complexity $O(|X|^{(k-1)/2})$, omitting factors that depend only on k. Another direction aims at obtaining more communication-efficient PSM schemes for functions of special interest. For example, a series of works [6,13,24,25,48] presented several "tailor-made" constructions for *symmetric* functions, whose outputs are independent of the order of inputs. PSM schemes have been used to obtain many other cryptographic primitives, e.g., secure computation with general interaction patterns [33], constant-round secure computation [35,37], and conditional disclosure of secrets [39,40]. The model of PSM schemes has also been generalized into different scenarios such as ad-hoc PSM [5,8] where only a subset of the parties actually send messages, and robust PSM (also known as non-interactive secure computation) [6,13] where an external party may corrupt some parties.

There are known lower bounds on the communication complexity of PSM. Applebaum et al. [2] proved a lower bound of $3 \log N - O(\log \log N)$ for the two-party case, and Ball and Randolph [4] showed a lower bound that is roughly $k^2 \log N$ for k-party PSM when $k = \omega(\log \log N)$. Tighter upper and lower bounds are known for concrete functions with small input domains and/or a small number of parties, including the n-bit AND function [23,29,49], the multi-input equality function [49], and the majority function [49].

A secret sharing scheme [14,47] is a cryptographic primitive, which divides a secret x into k shares in such a way that x can be recovered from any authorized set of shares while no information on x is revealed from any unauthorized set of shares. The collection of all authorized sets, namely sets of shares that can recover a secret, is called its access structure. A secret sharing scheme is called ideal if each share is taken from the same domain as secrets [15]. Since the size of each share cannot be smaller than the secret size [38], ideal secret sharing schemes necessarily achieve the optimal share size. A series of works have shown that several classes of access structures admit ideal secret sharing schemes [11,18,26–28,44,50], and found interesting connections to combinatorics and information theory [10, 16, 41]. Despite all these progresses, the exact characterization of access structures admitting ideal secret sharing schemes is a longstanding open problem.

2 Overview of Our Techniques

In this section, we provide an overview of our techniques. More detailed descriptions and formal proofs are given in the following sections.

2.1 Ideal PSM: Definition and Characterization

In a (two-party) PSM scheme, each of two parties has common randomness $r \in R$ and an input $x_i \in X_i$. Then, each party sends a single message $m_i \in M_i$ to an external party, from which he learns $f(x_0, x_1)$ for a public function $f : X_0 \times X_1 \to Y$. The PSM scheme specifies a randomized algorithm Gen to sample r, message functions $\mathsf{Msg}_i : X_i \times R_i \to M_i$ that computes m_i from (x_i, r), and an evaluation function $\mathsf{Eval} : M_0 \times M_1 \to Y$ that outputs $f(x_0, x_1)$ from (m_0, m_1). The privacy requirement is that the distribution of messages does not leak any information on the inputs other than $f(x_0, x_1)$. We say that the PSM scheme is *ideal* if parties' messages are taken from the same domain as inputs, i.e., $M_i = X_i$. A function f is called an *ideal PSM function* if there exists an ideal PSM scheme for f. The communication complexity of ideal PSM schemes cannot be reduced further if f is non-degenerate, by which we mean that its truth-table has no identical rows or columns. Throughout the paper, we only consider non-degenerate functions as the computation of degenerate functions is reduced to the computation of a smaller non-degenerate function.

We introduce a special class of ideal PSM schemes. As will be shown later, these schemes are canonical in the sense that every function admitting ideal PSM can be realized by some ideal PSM scheme in this class. Each of them is parameterized by a tuple $\mathcal{G} = (G_0, G_1, \psi)$, where G_i is a subgroup of the group $\mathcal{S}_{X_i}$ of all permutations over X_i, and ψ is an isomorphism from G_0 to G_1, that is, ψ maps a permutation belonging to G_0 to another permutation belonging to G_1. We define a group $\Gamma_{\mathcal{G}}$ as $\Gamma_{\mathcal{G}} = \{(\pi_0, \pi_1) \in G_0 \times G_1 : \pi_1 = \psi(\pi_0)\}$. Then, $\Gamma_{\mathcal{G}}$ naturally acts on $X_0 \times X_1$ as it is a subgroup of $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$, and thus specifies the set of *orbits*, $Y_{\mathcal{G}} = \{\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}$, where $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) = \{(\pi_0(x_0), \pi_1(x_1)) : (\pi_0, \pi_1) \in \Gamma_{\mathcal{G}}\}$. We define $f_{\mathcal{G}} : X_0 \times X_1 \to Y_{\mathcal{G}}$ as a function that maps each input to its orbit, namely

$$f_{\mathcal{G}}(x_0, x_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1).$$

We can construct an ideal PSM scheme $\Pi_{\mathcal{G}}$ for $f_{\mathcal{G}}$ as follows:

- Gen samples a permutation $\pi = (\pi_0, \pi_1)$ chosen from $\Gamma_{\mathcal{G}}$ uniformly at random.
- Msg_i applies the permutation π to an input x_i, namely $\mathsf{Msg}_i(x_i, \pi) = \pi_i(x_i)$.
- Eval outputs the orbit of messages (m_0, m_1), namely $\mathsf{Eval}(m_0, m_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(m_0, m_1)$.

Since an orbit of (x_0, x_1) is invariant under a permutation $\pi \in \Gamma_{\mathcal{G}}$, $\Pi_{\mathcal{G}}$ correctly computes $f_{\mathcal{G}}$. Furthermore, if $f_{\mathcal{G}}(x_0, x_1) = f_{\mathcal{G}}(x'_0, x'_1)$, then (x_0, x_1) and (x'_0, x'_1) are mapped to each other by some permutation $\pi \in \Gamma_{\mathcal{G}}$ as their orbits are identical, implying the privacy of $\Pi_{\mathcal{G}}$.

We show that every ideal PSM function $f : X_0 \times X_1 \to Y$ is equivalent (that is, identical up to permutations of inputs and outputs) to $f_{\mathcal{G}}$ for some $\mathcal{G}$. We obtain this characterization by proving that the following conditions are equivalent:

1. f is an ideal PSM function.
2. For any two inputs $(x_0, x_1), (x'_0, x'_1)$ such that $f(x_0, x_1) = f(x'_0, x'_1)$, there exists a pair of permutations $\pi = (\pi_0, \pi_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ such that
   - $\pi(x_0, x_1) = (\pi_0(x_0), \pi_1(x_1)) = (x'_0, x'_1)$.
   - $\pi \in I_f$, i.e., the values of f are invariant under π.

3. f is equivalent to $f_{\mathcal{G}}$ for some $\mathcal{G}$.

We have already seen the implication $3 \Rightarrow 1$.

To prove the implication $1 \Rightarrow 2$, we observe that since $M_i = X_i$ in an ideal PSM scheme, the message function $\mathsf{Msg}_i(\cdot, r)$ induces a permutation π_r^i over X_i for a random string $r \in R$. We construct S as a group generated by all such permutations (π_r^0, π_r^1)'s. We also observe that an ideal PSM scheme for f can be transformed into an ideal PSM scheme whose evaluation function is identical to f, namely $\mathsf{Eval}(m_0, m_1) = f(m_0, m_1)$. Then, the correctness property ensures that any permutation in S preserves the outputs of f since $f(\pi_r^0(x_0), \pi_r^1(x_1)) = \mathsf{Eval}(\mathsf{Msg}_0(x_0, r), \mathsf{Msg}_1(x_1, r)) = f(x_0, x_1)$. Furthermore, the privacy property ensures that if $f(x_0, x_1) = f(x_0', x_1')$, then there exist random strings $r, r' \in R$ such that $(\mathsf{Msg}_0(x_0, r), \mathsf{Msg}_1(x_1, r)) = (\mathsf{Msg}_0(x_0', r'), \mathsf{Msg}_1(x_1', r'))$. This implies that there exists a permutation $\pi \in S$ that maps (x_0, x_1) to (x_0', x_1').

To prove the implication $2 \Rightarrow 3$, we show a one-to-one correspondence between the set of orbits under the action of I_f and the range of f, that is,

$$\{\mathrm{Orb}_{I_f}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\} \xleftrightarrow{\text{1-to-1}} \{f(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}.$$

Indeed, if $\mathrm{Orb}_{I_f}(x_0, x_1) = \mathrm{Orb}_{I_f}(x_0', x_1')$, then the values of f on them are identical since f is invariant under any permutation in I_f. Conversely, if $f(x_0, x_1) = f(x_0', x_1')$, then there exists a permutation $\pi \in I_f$ that maps (x_0, x_1) to (x_0', x_1'), and hence their orbits are identical. Furthermore, we can construct a tuple $\mathcal{G} = (G_0, G_1, \psi)$ such that $\Gamma_{\mathcal{G}} = I_f$. Indeed, if f is non-degenerate, then any permutation π_0 appearing in the first component of I_f uniquely determines its counterpart π_1 such that $(\pi_0, \pi_1) \in I_f$. In particular, I_f, $\mathsf{pr}_0(I_f)$, and $\mathsf{pr}_1(I_f)$ are isomorphic to each other, where $\mathsf{pr}_i : \mathcal{S}_{X_0} \times \mathcal{S}_{X_1} \to \mathcal{S}_{X_i}$ is a projection map. We set $G_0 = \mathsf{pr}_0(I_f)$, $G_1 = \mathsf{pr}_1(I_f)$, and $\psi = \mathsf{pr}_1 \circ \mathsf{pr}_0^{-1}$. Note that G_i forms a group as pr_i is a homomorphism. Then, we have $I_f = \Gamma_{\mathcal{G}}$ where $\mathcal{G} = (G_0, G_1, \psi)$. Therefore, we obtain that $f_{\mathcal{G}}(x_0, x_1) = \mathrm{Orb}_{I_f}(x_0, x_1)$ and hence $f_{\mathcal{G}}$ is equivalent to f.

As a demonstration, we consider the sum function $f(x_0, x_1) = x_0 + x_1$ over an abelian group X. For $r \in X$, let $\sigma_r : X \to X$ denote a permutation that shifts each element $x \in X$ by r. It is easy to verify that if $f(x_0', x_1') = f(x_0, x_1)$, then $(x_0', x_1') = (\sigma_r(x_0), \sigma_{-r}(x_1))$ and $(\sigma_r, \sigma_{-r}) \in I_f$, where $r := x_0' - x_0$. Thus, our characterization ensures that f is an ideal PSM function. Note that f is equivalent to $f_{\mathcal{G}} = f_{(G_0, G_1, \psi)}$ where both G_0 and G_1 are the set consisting of all σ_r's, and $\psi : G_0 \to G_1$ is defined as $\psi(\sigma_r) = \sigma_{-r}$. Interestingly, this argument does not require presenting any explicit PSM protocol for f.

2.2 Enumeration of Ideal PSM Functions

The above characterization reduces the enumeration of ideal PSM functions $f : X_0 \times X_1 \to Y$ to the enumeration of all tuples $\mathcal{G}$'s. For ease of exposition, we assume that $|X_0| = |X_1| =: N$. See Section 5 for a general case.

First, we provide asymptotic upper bounds on the total number of such functions. Here, we focus on connected functions, that is, there exists no partition of $X_0 \times X_1$ into rectangles each of which corresponds to disjoint sets of values of f. In the boolean case $Y = \{0, 1\}$, this assumption excludes only a few trivial functions and does not affect the asymptotic results. Thanks to our characterization, we can obtain an upper bound by counting the number of all $\mathcal{G} = (G_0, G_1, \psi)$, where G_i is a subgroup of $\mathcal{S}_{X_i}$ and ψ is an isomorphism between G_0 and G_1. However, since the total number of subgroups of $\mathcal{S}_N$ is $2^{\Theta(N^2)}$ [45], naively enumerating all subgroups cannot provide a non-trivial upper bound beyond the trivial bound of 2^{N^2}, which is the number of all functions with domain $X_0 \times X_1$. To remove duplicate counts of $\mathcal{G}$'s that correspond to equivalent functions, we first observe that if $\mathcal{G}$ and $\mathcal{G}'$ are conjugate, then the corresponding functions $f_{\mathcal{G}}$ and $f_{\mathcal{G}'}$ are equivalent. Hence, it suffices to enumerate only conjugacy classes. As a more effective technique, we consider a minimal subgroup $H_{\mathcal{G}}$ of $\Gamma_{\mathcal{G}}$ that induces the same set of orbits as $\Gamma_{\mathcal{G}}$. Then, $H_{\mathcal{G}}$ uniquely determines $f_{\mathcal{G}}$, that is, $f_{\mathcal{G}}$ and $f_{\mathcal{G}'}$ are equivalent if $H_{\mathcal{G}} = H_{\mathcal{G}'}$. It thus suffices to bound the number of all such subgroups $H_{\mathcal{G}}$'s. A key observation is that $H_{\mathcal{G}}$ belongs to a special class of permutation groups called *orbit-minimal groups*. It is shown in [45] that any orbit-minimal group over a set X is generated by only $\log |X|$ permutations. As a result, any $H_{\mathcal{G}}$ is generated by at most $\log |X_0 \times X_1| = \log(N^2)$ pairs of permutations in $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$. We thus obtain a non-trivial upper bound on the number of ideal PSM functions as

$$\binom{|\mathcal{S}_{X_0}| \cdot |\mathcal{S}_{X_1}|}{\log |X_0 \times X_1|} = (N!)^{O(\log(N^2))} = 2^{O(N \log^2 N)}.$$

Furthermore, we obtain a tighter upper bound in the special case where both $|X_0|$ and $|X_1|$ are primes. We observe that it suffices to enumerate $\mathcal{G} = (G_0, G_1, \psi)$ such that G_0 and G_1 are transitive groups, when connected functions are considered. For a prime N, the total number of transitive subgroups of $\mathcal{S}_N$ is at most $2^{O(\log N)}$ [46]. Moreover, we show that the possible types of transitive groups relevant to our setting are limited, and in particular, their sizes are upper bounded by $2^{O(\log^2 N)}$. Consequently, in this special case, we obtain the following bound:

$$\sum_{G_0, G_1 : \text{transitive}} \binom{|G_0| \cdot |G_1|}{\log |X_0 \times X_1|} = 2^{O(\log N)} |G_0|^{\log(N^2)} |G_1|^{\log(N^2)} = 2^{O(\log^3 N)}.$$

Next, we provide a complete list of ideal PSM functions $f : X_0 \times X_1 \to Y$ for small domains. We use an open software package called GAP to enumerate all the conjugacy classes of $\mathcal{G}$. For a general range Y, we obtain a complete list up to $\max\{|X_0|, |X_1|\} \leq 15$ except for the case of $|X_0| = |X_1| = 12$. In the boolean case $Y = \{0, 1\}$, we devise a more computationally efficient method to enumerate $\mathcal{G}$. As a result, we obtain a list of boolean functions admitting ideal PSM up to $\max\{|X_0|, |X_1|\} \leq 23$. Specifically, if f is an ideal PSM function, i.e., $f = f_{\mathcal{G}}$ for some $\mathcal{G} = (G_0, G_1, \psi)$, then the truth-table T_f, viewed as a $|X_0|$-by-$|X_1|$ matrix, satisfies the following property: If $v \in \{0,1\}^{|X_1|}$ is a column of T_f, then so is a permuted vector $\pi_0(v)$ for any $\pi_0 \in G_0$. Conversely, for any pair of columns v, v' of T_f, there exists a permutation $\pi_0 \in G_0$ such that $v' = \pi_0(v)$. Leveraging this property, we can enumerate ideal PSM functions as follows: For each transitive group G_0, we construct a graph $\mathfrak{G}_{G_0} = (V_{G_0}, E_{G_0})$ where

$$V_{G_0} = \{0,1\}^n \text{ and } E_{G_0} = \{(v, v') \in V_{G_0} \times V_{G_0} : \exists \pi_0 \in G_0, v' = \pi_0(v)\}.$$

From the above property, the truth-table of any $f_{\mathcal{G}}$ with $\mathcal{G} = (G_0, G_1, \psi)$ corresponds to some connected component of $\mathfrak{G}_{G_0}$. We then collect the connected components of $\mathfrak{G}_{G_0}$ for all G_0's, which contain all desired functions.

2.3 Infinite Families of Ideal PSM Functions

We present several infinite families of ideal PSM functions. The simplest family is the *group product*, which computes

$$f((x_i)_{i \in S_0}, (y_i)_{i \in S_1}) = \prod_{i \in S_0 \cup S_1} z_i,$$

where each x_i and y_i are taken from a possibly non-abelian group G and we set $z_i = x_i$ for $i \in S_0$ and $z_i = y_i$ for $i \in S_1$. Feige et al. [29] constructed an ideal PSM scheme for this function by randomizing a sequence of group elements while preserving their product.

Next, we show that a function computing private set intersection cardinality admits ideal PSM. Specifically, for parameters $n, d_0, d_1 \in \mathbb{N}$, we define

$$\mathsf{PSIC}_{n,(d_0,d_1)}(A_0, A_1) = |A_0 \cap A_1|,$$

where A_i is taken from the set of all d_i-sized subsets of a common set U of n elements. To see that $\mathsf{PSIC}_{n,(d_0,d_1)}$ is an ideal PSM function, suppose that $|A_0 \cap A_1| = |A'_0 \cap A'_1|$ and let $B = A_0 \cap A_1$ and $B' = A'_0 \cap A'_1$. Then, we can choose a permutation π on U such that $\pi(A_0) = A'_0$ and $\pi(A_1) = A'_1$, and apply our characterization result. The existence of such π is guaranteed by the fact that both $A_0 \cup A_1$ and $A'_0 \cup A'_1$ admit partitions whose corresponding blocks have the same sizes, that is, $|A_0 \setminus B| = |A'_0 \setminus B'|$, $|A_1 \setminus B| = |A'_1 \setminus B'|$, and $|B| = |B'|$. In particular, $\mathsf{PSIC}_{n,(1,1)}$ is equivalent to the functionality of testing whether two given elements are identical, which implies that the equality function is an ideal PSM function.

Finally, we introduce a class of functions that generalizes the index function [29, 39]. Let H and G be abelian groups and let $\mathcal{F}$ be any set of families $\phi : H \to G$ that are closed under sums (that is, $\phi \pm \phi' \in \mathcal{F}$ for any $\phi, \phi' \in \mathcal{F}$) and are closed under input shifts (that is, $\phi_s \in \mathcal{F}$ for any $\phi \in \mathcal{F}, s \in H$, where $\phi_s(x) := \phi(x - s)$). We set $X_0 = H \times G$ and $X_1 = \mathcal{F}$, and define a function $f : (H \times G) \times \mathcal{F} \to G$ as

$$f((x, r), \phi) = \phi(x) - r.$$

This class of functions includes the original index function considered in [29], an augmented form of the inner product, and multi-linear polynomials as special instances. To show that these functions admit ideal PSM, we define two families of permutations over $X_0 \times X_1$:

- $\pi^s : ((x, r), \phi) \mapsto ((x + s, r), \phi_s)$ for any $s \in H$.
- $\sigma^\tau : ((x, r), \phi) \mapsto ((x, r + \tau(x)), \phi + \tau)$ for any $\tau \in \mathcal{F}$.

Intuitively, the first family of permutations allows us to freely move Alice's input (x, r) to any (x', r) while preserving the outputs of f. The second family allows us to freely move Bob's input ϕ to any ϕ'. Thus, for any pair of inputs $(x_0, x_1), (x_0', x_1')$ with $f(x_0, x_1) = f(x_0', x_1')$, we can find a permutation that maps (x_0, x_1) to (x_0', x_1') preserving the outputs of f and apply our characterization result.

2.4 Communication-efficient and Simplified PSM Schemes

In general, if we obtain a PSM scheme for a function $F : X_0' \times X_1' \to Y$, then the scheme naturally induces a PSM scheme for any function $f : X_0 \times X_1 \to Y$ that can be embedded into F, simply by restricting the input domains. Here, by saying that f is *embedded* into F, we mean that there are injections $\tau_i : X_i \to X_i'$ such that $f(x_0, x_1) = F(\tau_0(x_0), \tau_1(x_1))$ for all $(x_0, x_1) \in X_0 \times X_1$. Thus, the ideal PSM schemes described above particularly induce PSM schemes for functions that are embedded into the corresponding functions. We refer to such a construction template of PSM as the "embedding-to-ideal-PSM" construction. Interestingly, most of the existing PSM schemes achieving the state-of-the-art communication complexity in various computation models follow the embedding-to-ideal-PSM template. For example, the best known schemes for branching programs [29] and arithmetic formulas [19,34] are obtained from the ideal PSM scheme for the group product. The nearly optimal scheme for polynomials [39] is obtained from the ideal PSM scheme for a generalized index function, where $\mathcal{F}$ is the set of all multi-linear polynomials.

By embedding target functions to ideal PSM functions, we obtain novel communication-efficient PSM schemes for functions with sparse/dense or low-rank truth-tables. First, we consider a function $f : X_0 \times X_1 \to \{0, 1\}$ whose truth-table T_f, viewed as a $|X_0|$-by-$|X_1|$ matrix, contains at most d ones in each row for a small $d \ll N_1 := |X_1|$. By appending each row with an appropriate number of ones, we can embed f into a function $f' : X_0 \times X_1' \to \{0, 1\}$ whose truth-table contains exactly d ones in each row, where $N_1' := |X_1'| = N_1 + d$. Note that the truth-table of $\mathsf{PSIC}_{N_1',(d,1)}$ is a $\binom{N_1'}{d}$-by-N_1' matrix, which consists of all row vectors of weight d. We can then embed f' (and hence f) into $\mathsf{PSIC}_{N_1',(d,1)}$. The ideal PSM scheme for $\mathsf{PSIC}_{N_1',(d,1)}$ induces a PSM scheme for f with communication complexity $O(\log \binom{N_1'}{d}) = O(d \log(N_1 + d))$. Prior to this work, the only general construction in [7] can apply in this setting which results in communication complexity of $O(\sqrt{N_1})$. Thus, our construction achieves a strict improvement when $d = o(\sqrt{N_1}/\log N_1)$. Note that these schemes also apply to functions such that the rows (or the columns) are dense, that is, they contain at least $n - d$ ones for a small d.

Next, Narayanan et al. [43] showed a PSM scheme for a function $f : X_0 \times X_1 \to Y$ whose communication complexity is $O(k \log |Y|)$, where k is the tiling number of f. Here, the tiling number refers to the smallest number of disjoint monochromatic rectangles covering the truth-table T_f. Let q be the smallest prime power larger than $|Y|$ (we can choose q so that $q = O(|Y|)$). If the rank of the truth-table T_f is at most r over $\mathbb{F}_q$, then we can decompose $T_f = \mathbf{M}_0\mathbf{M}_1^\top$ where $\mathbf{M}_i \in \mathbb{F}_q^{|X_i| \times r}$. Thus, by defining an injection

$$\tau_i : X_i \ni x_i \mapsto \mathbf{M}_i[x_i] \in \mathbb{F}_q^r,$$

where $\mathbf{M}_i[x]$ denotes the x-th row of $\mathbf{M}_i$, we can embed f into the inner product function over $\mathbb{F}_q^r$. The inner product is a special case of the generalized index function and hence admits ideal PSM. As a result, we obtain a PSM scheme for f with communication complexity $O(r \log |Y|)$. Since it holds that $k \geq r$ for any f, our construction achieves a more refined communication complexity than [43].

Finally, Beimel et al. [7] showed a PSM scheme for a general function $f : X \times X \to \{0,1\}$ with communication complexity $O(\sqrt{N})$, where $X = \{0,1,\dots,N-1\}$. According to the description in [3], the more precise communication complexity is at least $12\sqrt{N}$. In contrast, we show that any function f can be embedded into an ideal PSM function $G_f : X' \times X' \to \{0,1\}$ with $\log |X'| = 4\sqrt{N}+1$. Precisely, we consider a multi-linear function $\hat{f} : (\{0,1\}^{\sqrt{N}})^4 \to \{0,1\}$ such that

$$\hat{f}(\mathbf{e}_{i_0}, \mathbf{e}_{j_0}, \mathbf{e}_{i_1}, \mathbf{e}_{j_1}) = f(i_0\sqrt{N} + j_0, i_1\sqrt{N} + j_1)$$

for any $i_0, j_0, i_1, j_1 \in \{0,1,\dots,\sqrt{N}-1\}$, where $\mathbf{e}_i$ denotes a one-hot vector whose i-th element is one. Then, we set $X' = (\{0,1\}^{\sqrt{N}})^4 \times \{0,1\}$ and define a function $G_f : X' \times X' \to \{0,1\}$ as

$$G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) = \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{x}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{x}_1, \mathbf{j}_1 \rangle \oplus \langle \mathbf{y}_0, \mathbf{i}_0 \rangle \oplus \langle \mathbf{y}_1, \mathbf{i}_1 \rangle \oplus a \oplus b$$

for any $\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, \mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1 \in \{0,1\}^{\sqrt{N}}$ and $a, b \in \{0,1\}$. To show that G_f admits ideal PSM, we construct the following families of permutations over $X' \times X'$ under which the outputs of G_f are invariant:

- For any $\mathbf{t}_0, \mathbf{t}_1 \in \{0,1\}^{\sqrt{N}}$, $\pi^{\mathbf{t}_0,\mathbf{t}_1}$ freely shifts part of Alice's input $(\mathbf{x}_0, \mathbf{x}_1)$ by $(\mathbf{t}_0, \mathbf{t}_1)$.
- For any $\mathbf{s}_0, \mathbf{s}_1 \in \{0,1\}^{\sqrt{N}}$, $\sigma^{\mathbf{s}_0,\mathbf{s}_1}$ freely shifts part of Bob's input $(\mathbf{y}_0, \mathbf{y}_1)$ by $(\mathbf{s}_0, \mathbf{s}_1)$.
- For any $\mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1 \in \{0,1\}^{\sqrt{N}}$, $\tau^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1}$ freely shifts part of Alice's and Bob's inputs, $(\mathbf{i}_0, \mathbf{i}_1)$ and $(\mathbf{j}_0, \mathbf{j}_1)$, by $(\mathbf{p}_0, \mathbf{p}_1)$ and $(\mathbf{q}_0, \mathbf{q}_1)$, respectively.
- For any $r \in \{0,1\}$, ρ^r maps part of Alice's and Bob's inputs (a, b) to $(a \oplus r, b \oplus r)$.

Thus, we can map an input to any other input by composing permutations from the above families as long as their corresponding outputs of G_f are equal, which shows the existence of ideal PSM for G_f. Since the communication complexity of the resulting ideal PSM scheme is $\log |X' \times X'| = 8\sqrt{N}+2$, our construction improves the constant factor in the dominant term of the best known communication complexity. An additional benefit is that Alice and Bob can compute their messages simply by applying a composition of the permutations defined above to their inputs. As a result, the message functions of our scheme can be expressed in closed form and simplifies the hierarchical design of [3, 7], which relies on nested invocations of PSM schemes for smaller functions.

3 Preliminaries

This section summarizes basic notations used in the paper. For $n \in \mathbb{N}$, we define $[n] = \{0,1,2,\dots,n-1\}$. For a set X, let $\binom{X}{k}$ denote the set of all k-sized subsets of X. We write $x \leftarrow_\$ X$ if an element x is sampled uniformly at random from a set X. The statistical distance between two random variables X, Y over a set U is defined as $\mathsf{SD}(X,Y) = (1/2)\sum_{a \in U} |\Pr[X = a] - \Pr[Y = a]|$. For two vectors $\mathbf{u}, \mathbf{v}$, we denote their inner product by $\langle \mathbf{u}, \mathbf{v} \rangle$ and their tensor (or Kronecker product) by $\mathbf{u} \otimes \mathbf{v}$. Let $\mathbb{F}_q$ denote a finite field of size q.

3.1 Permutation Groups

If H is a subgroup of a group G, then we write $H \leq G$. We denote the group of all permutations over a finite set X by $\mathcal{S}_X$. If $|X| = N$, then we also denote it by $\mathcal{S}_N$ and call it the permutation group of degree N. We denote the inverse of a permutation π by π^{-1} and the composition of π and π' by $\pi' \circ \pi$. We say that two subgroups $G, H \leq \mathcal{S}_X$ are *conjugate* if there exists a permutation $\tau \in \mathcal{S}_X$ such that

$G = \tau^{-1}H\tau = \{\tau^{-1}h\tau : h \in H\}$. We say that G is generated by a subset $S \subseteq G$ if every element of G can be expressed by a composition of finitely many elements of S or their inverses.

Each permutation $\pi \in \mathcal{S}_X$ naturally acts on the set X. If $n = |X|$, then π also acts on the set Y^n of n-dimensional vectors over a set Y: for each $v = (v_x)_{x\in X} \in Y^n$, define $\pi(v) = (v'_x)_{x\in X} \in Y^n$ as $v'_{\pi(x)} = v_x$ for all $x \in X$. For a subset $S \subseteq \mathcal{S}_X$, we define the *orbit* of $x \in X$ under the action of S on X by $\mathrm{Orb}_S(x) = \{\pi(x) : \pi \in S\}$. We say that $G \leq \mathcal{S}_X$ is *transitive* if for any $x, x' \in X$, there exists $\pi \in G$ such that $\pi(x) = x'$, that is, the orbit of any $x \in X$ is equal to X.

Let X_0, X_1 be sets. We naturally identify $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ as a subgroup of $\mathcal{S}_{X_0\times X_1}$ via $\tau(x_0, x_1) = (\tau_0(x_0), \tau_1(x_1))$ where $\tau = (\tau_0, \tau_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ and $(x_0, x_1) \in X_0 \times X_1$. We denote a natural projection from $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ onto $\mathcal{S}_{X_i}$ by pr_i. For $S \subseteq \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$, we similarly define the orbit of (x_0, x_1) under the action of S on $X_0 \times X_1$, denoted by $\mathrm{Orb}_S(x_0, x_1)$.

3.2 Functions

Let $f : X_0 \times X_1 \to Y$ be a function. For subsets $A_0 \subseteq X_0$ and $A_1 \subseteq X_1$, we denote $f(A_0 \times A_1) = \{f(x_0, x_1) : (x_0, x_1) \in A_0 \times A_1\}$ and denote $\mathrm{Range}(f) = f(X_0 \times X_1)$. We also denote $f^{-1}(y) = \{(x_0, x_1) \in X_0 \times X_1 : f(x_0, x_1) = y\}$ for $y \in \mathrm{Range}(f)$.

We define an equivalence relation $\simeq$ on the set of all functions whose domain is $X_0 \times X_1$ as follows:

$$f \simeq g \quad \stackrel{\text{def}}{\iff} \quad \forall (x_0, x_1) \in X_0 \times X_1,\ \sigma(f(\tau_0(x_0), \tau_1(x_1))) = g(x_0, x_1)$$

for some pair of permutations $(\tau_0, \tau_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ and some bijection $\sigma : \mathrm{Range}(f) \to \mathrm{Range}(g)$.

The truth-table of $f : X_0 \times X_1 \to Y$ is defined as a matrix T_f whose rows and columns are indexed by X_0 and X_1 and whose (x_0, x_1)-th entry $T_f[x_0, x_1]$ is $f(x_0, x_1)$ for any $(x_0, x_1) \in X_0 \times X_1$. For each $x_0 \in X_0$, we denote the row vector of T_f corresponding to x_0 by $T_f[x_0, *] \in Y^{|X_1|}$. Similarly, we denote the column vector of T_f corresponding to $x_1 \in X_1$ by $T_f[*, x_1] \in Y^{|X_0|}$.

We define the *NOT* of $f : X_0 \times X_1 \to \{0, 1\}$, denoted by $\overline{f}$, as a function $\overline{f} : X_0 \times X_1 \to \{0, 1\}$ such that $\overline{f}(x_0, x_1) = 1 - f(x_0, x_1)$ for any $(x_0, x_1) \in X_0 \times X_1$. We also define the *transpose* of f, denoted by $f^\top$, as a function $f^\top : X_1 \times X_0 \to \{0, 1\}$ such that $f^\top(x_1, x_0) = f(x_0, x_1)$ for any $(x_1, x_0) \in X_1 \times X_0$.

We say that a function $f : X_0 \times X_1 \to Y$ is *degenerate* if there exist $i \in \{0, 1\}$ and $x_i \neq x'_i \in X_i$ such that $f(x_i, x_{1-i}) = f(x'_i, x_{1-i})$ for all $x_{1-i} \in X_{1-i}$, where the inputs are supposed to be sorted according to the index. We say that f is *non-degenerate* if f is not degenerate.

We say that $f : X_0 \times X_1 \to Y$ is *disconnected* if there exist $i \in \{0, 1\}$ and a partition $X_i = A_i \cup B_i$ such that $f(A_i \times X_{1-i}) \cap f(B_i \times X_{1-i}) = \emptyset$. We say that f is *connected* if f is not disconnected. Note that if $Y = \{0, 1\}$, then non-degenerate disconnected functions are limited to those whose truth tables are (1 0) or its transpose, or functions that are equivalent to them. For $i \in \{0, 1\}$, we define a graph $\mathfrak{G}_{f,i} = (V, E)$ as

$$V = X_i \text{ and } E = \{(x_i, x'_i) \in X_i^2 : f(\{x_i\} \times X_{1-i}) \cap f(\{x'_i\} \times X_{1-i}) \neq \emptyset\}.$$

If f is a connected function, then $\mathfrak{G}_{f,i}$ is a connected graph. Indeed, if $\mathfrak{G}_{f,i} = (V, E)$ is not connected, then there exists a partition $V = A \cup B$ such that there is no edge between any pair (a, b) of vertices $a \in A$ and $b \in B$. Then, we have that $f(A \times X_{1-i}) \cap f(B \times X_{1-i}) = \emptyset$, implying that f is disconnected.

For $\pi = (\pi_0, \pi_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ and $x = (x_0, x_1) \in X_0 \times X_1$, we simply write $\pi(x) = (\pi_0(x_0), \pi_1(x_1))$ and $f(\pi(x)) = f(\pi_0(x_0), \pi_1(x_1))$. We define the *invariant group* of f, denoted by I_f, as the subgroup consisting of all pairs of permutations in $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ under which the outputs of f are invariant, i.e.,

$$I_f = \{\pi \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1} : f(\pi(x)) = f(x), \forall x \in X_0 \times X_1\}.$$

3.3 Private Simultaneous Messages

We introduce Private Simultaneous Messages (PSM) schemes. Each of two parties holds an input x_i and they both share a common string r sampled by a randomized algorithm Gen. Then, each party runs an algorithm Msg to compute a message m_i. They send the messages to an external party, who then runs an algorithm Eval to obtain $f(x_0, x_1)$. The privacy requirement is that the joint distribution of messages leaks no extra information. We provide the formal definition below.

Definition 3.1. *Let $f : X_0 \times X_1 \to Y$ be a function. A* Private Simultaneous Messages (PSM) *scheme Π for f is a tuple $(\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ of three algorithms, where*

- $\mathsf{Gen}() \to r$*: Gen is a randomized algorithm that takes no input and outputs $r \in R$ sampled from a probability distribution over a set R.*
- $\mathsf{Msg}(i, x_i, r)$*: Msg is a deterministic algorithm that takes $i \in \{0, 1\}$, $x_i \in X_i$, and $r \in R$ as input and outputs a message m_i.*
- $\mathsf{Eval}(m_0, m_1)$*: Eval is a deterministic algorithm that takes messages (m_0, m_1) as input and outputs $y \in Y$.*

satisfying the following properties:

Correctness. *For any $(x_0, x_1) \in X_0 \times X_1$, it holds that*

$$\Pr[\mathsf{Eval}(\mathsf{Msg}(0, x_0, r), \mathsf{Msg}(1, x_1, r)) = f(x_0, x_1)] = 1, \tag{1}$$

where the probability is taken over a random choice of $r \leftarrow \mathsf{Gen}()$.

Privacy. *For any input $(x_0, x_1) \in X_0 \times X_1$, let $\mathcal{D}_{(x_0,x_1)}$ denote the probability distribution of $(\mathsf{Msg}(i, x_i, r))_{i\in\{0,1\}}$ induced by $r \leftarrow \mathsf{Gen}()$. For any pair of inputs $(x_0, x_1), (x'_0, x'_1) \in X_0 \times X_1$ with $f(x_0, x_1) = f(x'_0, x'_1)$, it holds that*

$$\mathsf{SD}\left(\mathcal{D}_{(x_0,x_1)}, \mathcal{D}_{(x'_0,x'_1)}\right) = 0. \tag{2}$$

We call the set of all possible outcomes of Gen (i.e., R) the randomness space of Π. We call the set of all possible outcomes of $\mathsf{Msg}(i, \cdot, \cdot)$ the i-th message space and denote it by M_i. The communication complexity of Π is defined as $\log |M_0| + \log |M_1|$.

We can consider a weaker privacy requirement by replacing the condition (2) with the following condition:

$$\mathsf{SD}\left(\mathcal{D}_{(x_0,x_1)}, \mathcal{D}_{(x'_0,x'_1)}\right) < 1. \tag{3}$$

We refer to PSM schemes satisfying the perfect correctness (i.e., (1)) and the above weaker privacy (i.e., (3)) as *weakly private PSM* schemes.

If a PSM scheme for a function f is given, then it immediately implies a PSM scheme with the same communication complexity for any function f' that is equivalent to f. Therefore, throughout this paper, we do not distinguish between equivalent functions and rather focus on equivalence classes of functions.

Furthermore, we focus on PSM schemes for non-degenerate functions. Suppose that $f : X_0 \times X_1 \to Y$ is degenerate, and that there exist $x_0, x'_0 \in X_0$ such that $f(x_0, x_1) = f(x'_0, x_1)$ for any $x_1 \in X_1$. Then, any PSM scheme Π' for the restriction f' of f to $X_0 \setminus \{x'_0\} \times X_1$ can be extended to a PSM scheme Π for f simply by letting party 0 "reuse" his message for x_0 when his input is x'_0. The communication complexity of Π is the same as that of Π'.

4 Ideal PSM

In this section, we introduce the notion of ideal PSM schemes, in which messages are taken from the same domain as inputs. Ideal PSM schemes are necessarily communication-optimal as communication complexity must be at least the input length of a function. We begin by presenting the formal definition.

Definition 4.1. *We say that a PSM scheme* Π *for a function* $f : X_0 \times X_1 \to Y$ *is* ideal *if the message spaces satisfy*

$$M_0 = X_0 \text{ and } M_1 = X_1. \tag{4}$$

We say that a function f admits ideal PSM *or is an* ideal PSM function *if there exists an ideal PSM scheme for* f*.*

Suppose that Π is an ideal PSM scheme for a non-degenerate function f. Then, the communication complexity of Π cannot be reduced further. This is because otherwise, for some randomness r and some pair of inputs, say x_0, x_0', the corresponding messages coincide. Then, the perfect correctness of Π implies that $f(x_0, x_1) = f(x_0', x_1)$ for any x_1, which contradicts that f is non-degenerate.

Clearly, f is an ideal PSM function if and only if there exists a PSM scheme Π for f such that the bit-length of each message is equal to that of each input, i.e., $|M_i| = |X_i|$. Indeed, such a PSM scheme $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ implies an ideal PSM scheme $\Pi' = (\mathsf{Gen}', \mathsf{Msg}', \mathsf{Eval}')$ as follows: For each $i \in \{0,1\}$, fix a bijection $\sigma_i : M_i \to X_i$. Then, define $\mathsf{Gen}' = \mathsf{Gen}$, $\mathsf{Msg}'(i, x_i, r) = \sigma_i(\mathsf{Eval}(i, x_i, r))$, and $\mathsf{Eval}((m_i)_{i\in\{0,1\}}) = \mathsf{Eval}((\sigma_i^{-1}(m_i))_{i\in\{0,1\}})$. Thus, the class of ideal PSM functions is preserved if the condition (4) is replaced with $|M_i| = |X_i|$ for $i \in \{0,1\}$.

4.1 Canonical Ideal PSM Schemes

We introduce a special class of ideal PSM schemes induced by permutation groups. Jumping ahead, we will show that the set of ideal PSM functions is essentially limited to those realized by this special type of ideal PSM schemes. In that sense, we refer to these ideal schemes as "canonical."

For sets X_0, X_1, we define

$$\Psi_{X_0,X_1} = \{(G_0, G_1, \psi) : G_0 \le \mathcal{S}_{X_0}, G_1 \le \mathcal{S}_{X_1},\ \psi : G_0 \to G_1 \text{ is an isomorphism}\}.$$

For each $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$, we define a subgroup $\Gamma_\mathcal{G}$ of $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ as

$$\Gamma_\mathcal{G} = \{(\pi_0, \pi_1) : \pi_0 \in G_0, \pi_1 = \psi(\pi_0)\}.$$

and $Y_\mathcal{G} = \{\mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}$. Finally, we define a function $f_\mathcal{G} : X_0 \times X_1 \to Y_\mathcal{G}$ by

$$f_\mathcal{G}(x_0, x_1) = \mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0, x_1)$$

for all $(x_0, x_1) \in X_0 \times X_1$.

We can see that each function $f_\mathcal{G}$ admits ideal PSM.

Proposition 4.2. *For each* $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$*, there exists an ideal PSM scheme* $\Pi_\mathcal{G}$ *for the function* $f_\mathcal{G}$*.*

Proof. We construct $\Pi_\mathcal{G}$ as follows. The randomness generation algorithm Gen outputs $\pi = (\pi_0, \pi_1)$ chosen uniformly at random from $\Gamma_\mathcal{G}$. The message function Msg is defined as $\mathsf{Msg}(i, x_i, \pi) = \pi_i(x_i)$. The evaluation function Eval is defined as $\mathsf{Eval}(m_0, m_1) = \mathrm{Orb}_{\Gamma_\mathcal{G}}(m_0, m_1)$. By definition, $\Pi_\mathcal{G}$ is ideal.

We have that

$$\mathsf{Eval}(\mathsf{Msg}(0, x_0, \pi), \mathsf{Msg}(1, x_1, \pi)) = \mathrm{Orb}_{\Gamma_\mathcal{G}}(\pi_0(x_0), \pi_1(x_1)) = \mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0, x_1) = f_\mathcal{G}(x_0, x_1).$$

The second equality follows from $\pi = (\pi_0, \pi_1) \in \Gamma_\mathcal{G}$. Thus, the correctness of $\Pi_\mathcal{G}$ follows.

Let $(x_0, x_1), (x_0', x_1')$ be two inputs such that $f_\mathcal{G}(x_0, x_1) = f_\mathcal{G}(x_0', x_1')$. Then, $\mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0, x_1) = \mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0', x_1')$. It follows from the orbit-stabilizer theorem that the distribution of $(\mathsf{Msg}(0, x_0, \pi), \mathsf{Msg}(1, x_1, \pi)) = (\pi_0(x_0), \pi_1(x_1))$ induced by $\pi \leftarrow_\$ \Gamma_\mathcal{G}$ is a uniform distribution over the orbit $\mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0, x_1)$. Similarly, the distribution of $(\mathsf{Msg}(0, x_0', \pi'), \mathsf{Msg}(1, x_1', \pi'))$ induced by $\pi' \leftarrow_\$ \Gamma_\mathcal{G}$ is a uniform distribution over $\mathrm{Orb}_{\Gamma_\mathcal{G}}(x_0', x_1')$. Therefore, the distribution of $(\mathsf{Msg}(i, x_i, \pi))_{i\in\{0,1\}}$ is identical with that of $(\mathsf{Msg}(i, x_i', \pi'))_{i\in\{0,1\}}$, from which the privacy of $\Pi_\mathcal{G}$ follows. $\square$

4.2 Characterization

We show that the class of ideal PSM functions is essentially equal to those computed by the canonical ideal PSM schemes, namely the $f_{\mathcal{G}}$'s.

First, we prepare two lemmas.

Lemma 4.3. *Let $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ be a PSM scheme for a non-degenerate function $f : X_0 \times X_1 \to Y$ with randomness space R. Then, for any $r \in R$ and $i \in \{0,1\}$, the function $\mathsf{Msg}(i, \cdot, r) \colon X_i \to M_i$ is injective. In particular, if Π is ideal, then $\mathsf{Msg}(i, \cdot, r) \colon X_i \to X_i$ is a bijection.*

Proof. Suppose to the contrary that $\mathsf{Msg}(0, \cdot, r)$ is not injective. Then, there exist $x_0 \neq x_0' \in X_0$ such that $\mathsf{Msg}(0, x_0, r) = \mathsf{Msg}(0, x_0', r)(=: m_0)$. From the correctness requirement of PSM, we have

$$f(x_0, x_1) = \mathsf{Eval}(m_0, \mathsf{Msg}(1, x_1, r)) = f(x_0', x_1)$$

for any $x_1 \in X_1$. This contradicts that f is non-degenerate. From the similar argument, the function $\mathsf{Msg}(1, \cdot, r) \colon X_1 \to M_1$ is injective for any $r \in R$. $\square$

Lemma 4.4. *Assume that a non-degenerate function $f : X_0 \times X_1 \to Y$ admits ideal PSM. Then, there exists an ideal PSM scheme $\Pi' = (\mathsf{Gen}', \mathsf{Msg}', \mathsf{Eval}')$ for f such that $\mathsf{Eval}' = f$.*

Proof. Let Π be an ideal PSM for f. Let R be a randomness space of Π and $r_0 \in R$ be an arbitrarily fixed randomness. Note that $\mathsf{Msg}(i, \cdot, r_0) : X_i \to X_i$ is a bijection from Lemma 4.3. Let $\mu_i : X_i \to X_i$ be the inverse of $\mathsf{Msg}(i, \cdot, r_0)$. We define $\Pi' = (\mathsf{Gen}', \mathsf{Msg}', \mathsf{Eval}')$ as follows:

- $\mathsf{Gen}' = \mathsf{Gen}$.
- $\mathsf{Msg}'(i, x_i, r) = \mu_i(\mathsf{Msg}(i, x_i, r))$.
- $\mathsf{Eval}'(m_0, m_1) = f(m_0, m_1)$.

We prove that Π' is an ideal PSM scheme for f. From the correctness of Π, it holds that

$$\mathsf{Eval}(\mu_0^{-1}(x_0'), \mu_1^{-1}(x_1')) = \mathsf{Eval}(\mathsf{Msg}(0, x_0', r_0), \mathsf{Msg}(1, x_1', r_0)) = f(x_0', x_1')$$

for any $(x_0', x_1') \in X_0 \times X_1$. For any input (x_0, x_1) and randomness r, by substituting $x_i' = \mu_i(\mathsf{Msg}(i, x_i, r))$, we have that

$$\mathsf{Eval}(\mathsf{Msg}(0, x_0, r), \mathsf{Msg}(1, x_1, r)) = f(\mu_0(\mathsf{Msg}(0, x_0, r)), \mu_1(\mathsf{Msg}(1, x_1, r))).$$

Thus, the correctness of Π' follows as

$$\begin{aligned}
\mathsf{Eval}'(\mathsf{Msg}'(0, x_0, r), \mathsf{Msg}'(1, x_1, r)) &= f(\mu_0(\mathsf{Msg}(0, x_0, r)), \mu_1(\mathsf{Msg}(1, x_1, r))) \\
&= \mathsf{Eval}(\mathsf{Msg}(0, x_0, r), \mathsf{Msg}(1, x_1, r)) \\
&= f(x_0, x_1).
\end{aligned}$$

The distribution of $(\mathsf{Msg}'(i, x_i, r))_{i \in \{0,1\}} = (\mu_i(\mathsf{Msg}(i, x_i, r)))_{i \in \{0,1\}}$ only depends on the distribution of $(\mathsf{Msg}(i, x_i, r))_{i \in \{0,1\}}$ since μ_0 and μ_1 are deterministic functions. The latter distribution only depends on $f(x_0, x_1)$ from the privacy requirement of Π, and therefore the distribution of $(\mathsf{Msg}'(i, x_i, r))_{i \in \{0,1\}}$ also only depends on $f(x_0, x_1)$. Therefore, Π' is a PSM scheme for f. Also, Π' is ideal since the message spaces of Π' are equal to the input spaces. $\square$

Now, we show a characterization of ideal PSM functions.

Theorem 4.5. *Let $f : X_0 \times X_1 \to Y$ be a non-degenerate function. Then, the following conditions are equivalent:*

1. *f admits ideal PSM.*

2. *There exists $\mathcal{G} \in \Psi_{X_0,X_1}$ such that $f \simeq f_{\mathcal{G}}$.*

3. *It holds that $\mathrm{Orb}_{I_f}(x_0, x_1) = f^{-1}(f(x_0, x_1))$ for any $(x_0, x_1) \in X_0 \times X_1$.*

4. *There exists a subset $S \subseteq I_f$ such that $\mathrm{Orb}_S(x_0, x_1) = f^{-1}(f(x_0, x_1))$ for any $(x_0, x_1) \in X_0 \times X_1$.*

Proof. **2 ⇒ 1.** Let $\Pi_{\mathcal{G}} = (\mathsf{Gen}_{\mathcal{G}}, \mathsf{Msg}_{\mathcal{G}}, \mathsf{Eval}_{\mathcal{G}})$ be the ideal PSM scheme for $f_{\mathcal{G}}$ constructed in Proposition 4.2. Since $f \simeq f_{\mathcal{G}}$, there are a pair of permutations $\tau = (\tau_0, \tau_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ and a bijection $\sigma : Y \to Y_{\mathcal{G}}$ such that $\sigma(f(\tau(x))) = f_{\mathcal{G}}(x)$ for all $x = (x_0, x_1) \in X_0 \times X_1$.

It suffices to construct an ideal PSM scheme $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ for f. We define Π as follows:

- $\mathsf{Gen} = \mathsf{Gen}_{\mathcal{G}}$, that is, Gen outputs $\pi = (\pi_0, \pi_1) \leftarrow_{\$} \Gamma_{\mathcal{G}}$.
- $\mathsf{Msg}(i, x_i, \pi) = \mathsf{Msg}_{\mathcal{G}}(i, \tau_i^{-1}(x_i), \pi) = \pi_i(\tau_i^{-1}(x_i))$.
- $\mathsf{Eval}(m_0, m_1) = \sigma^{-1}(\mathsf{Eval}_{\mathcal{G}}(m_0, m_1))$.

The correctness follows since

$$\begin{aligned}\mathsf{Eval}(\mathsf{Msg}(0, x_0, \pi), \mathsf{Msg}(1, x_1, \pi)) &= \sigma^{-1}(\mathsf{Eval}_{\mathcal{G}}(\mathsf{Msg}_{\mathcal{G}}(0, \tau_0^{-1}(x_0), \pi), \mathsf{Msg}_{\mathcal{G}}(1, \tau_1^{-1}(x_1), \pi)) \\ &= \sigma^{-1}(f_{\mathcal{G}}(\tau_0^{-1}(x_0), \tau_1^{-1}(x_1))) \\ &= f(x_0, x_1).\end{aligned}$$

The privacy of Π is implied by that of $\Pi_{\mathcal{G}}$ since $f(x_0, x_1) = f(x_0', x_1')$ if and only if $f_{\mathcal{G}}(\tau_0^{-1}(x_0), \tau_1^{-1}(x_1)) = f_{\mathcal{G}}(\tau_0^{-1}(x_0'), \tau_1^{-1}(x_1'))$. Clearly, Π is ideal.

1 ⇒ 4. Let Π be an ideal PSM scheme for f obtained via Lemma 4.4. Let R be the randomness space of Π. For each $r \in R$ and $i \in \{0, 1\}$, we define $\pi_r^i : X_i \to X_i$ as $\pi_r^i(x_i) = \mathsf{Msg}(i, x_i, r)$. From Lemma 4.3, π_r^i is a permutation on X_i. Let S be a subgroup of $\mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ generated by $\{(\pi_r^0, \pi_r^1) : r \in R\}$.

First, we prove that $S \subseteq I_f$. Let $(x_0, x_1) \in X_0 \times X_1$ and $(\pi_0, \pi_1) \in S$. Since S is generated by $\{(\pi_r^0, \pi_r^1) : r \in R\}$, there exists a sequence $(r_1, \ldots, r_m)$ such that

$$(\pi_0, \pi_1) = (\pi_{r_1}^0, \pi_{r_1}^1) \circ \cdots \circ (\pi_{r_m}^0, \pi_{r_m}^1). \tag{5}$$

Note that since S is a finite group, for any $(\pi_0, \pi_1) \in S$, there exists $k > 0$ such that $(\pi_0, \pi_1)^{-1} = (\pi_0, \pi_1)^k$. Hence, any $(\pi_0, \pi_1) \in S$ can be represented as in Eq. (5). Then, from the correctness of Π, we have

$$f(x_0, x_1) = f(\pi_{r_m}^0(x_0), \pi_{r_m}^1(x_1)) = f(\pi_{r_{m-1}}^0 \circ \pi_{r_m}^0(x_0), \pi_{r_{m-1}}^1 \circ \pi_{r_m}^1(x_1)) = \cdots = f(\pi_0(x_0), \pi_1(x_1)),$$

for all $(x_0, x_1) \in X_0 \times X_1$.

Next, we prove that $\mathrm{Orb}_S(x_0, x_1) = f^{-1}(f(x_0, x_1))$ for any $(x_0, x_1) \in X_0 \times X_1$. Since we have shown $S \subseteq I_f$, it holds that $\mathrm{Orb}_S(x_0, x_1) \subseteq f^{-1}(f(x_0, x_1))$. It remains to show that for any $(x_0', x_1') \in f^{-1}(f(x_0, x_1))$, there exists $(\pi_0, \pi_1) \in S$ such that $\pi_0(x_0) = x_0'$ and $\pi_1(x_1) = x_1'$. From the privacy requirement of Π, the distribution of $(\pi_r^0(x_0), \pi_r^1(x_1))$ induced by $r \leftarrow_{\$} R$ is identical with that of $(\pi_{r'}^0(x_0'), \pi_{r'}^1(x_1'))$ induced by $r' \leftarrow_{\$} R$, since $f(x_0, x_1) = f(x_0', x_1')$. This particularly implies that there exist $r, r' \in R$ such that $(\pi_r^0(x_0), \pi_r^1(x_1)) = (\pi_{r'}^0(x_0'), \pi_{r'}^1(x_1'))$. By the definition of S,

$$(\pi_0, \pi_1) := (\pi_{r'}^0, \pi_{r'}^1)^{-1} \circ (\pi_r^0, \pi_r^1)$$

belongs to S and $(\pi_0(x_0), \pi_1(x_1)) = (x_0', x_1')$.

4 ⇒ 3. Let $(x_0, x_1) \in X_0 \times X_1$. By the definition of I_f, it holds that $\mathrm{Orb}_{I_f}(x_0, x_1) \subseteq f^{-1}(f(x_0, x_1))$. The converse inclusion follows from $f^{-1}(f(x_0, x_1)) = \mathrm{Orb}_S(x_0, x_1) \subseteq \mathrm{Orb}_{I_f}(x_0, x_1)$ as $S \subseteq I_f$.

3 ⇒ 2. Define $G_i = \mathsf{pr}_i(I_f)$ for $i \in \{0, 1\}$.

First, we prove that the restriction of the projection pr_i to I_f is an isomorphism, i.e., G_i and I_f are isomorphic. Since pr_i is surjective, it is enough to show that it is injective. Suppose on the contrary that there exist $\pi_0 \in G_0$ and $\pi_1 \neq \pi_1' \in G_1$ such that $(\pi_0, \pi_1) \in I_f$ and $(\pi_0, \pi_1') \in I_f$. Then, there exists $x_1 \in X_1$ such that $\pi_1(x_1) \neq \pi_1'(x_1)$. For any $x_0 \in X_0$, we have

$$f(\pi_0(x_0), \pi_1(x_1)) = f(x_0, x_1) = f(\pi_0(x_0), \pi_1'(x_1)).$$

Since π_0 is a bijection, this implies that for any $x_0 \in X_0$, we have $f(x_0, \pi_1(x_1)) = f(x_0, \pi_1'(x_1))$. This contradicts that f is non-degenerate. Thus, the projection $\mathsf{pr}_i : I_f \to G_i$ is an injection and hence is an isomorphism. In summary, G_0, G_1 and I_f are isomorphic to each other. Let ψ be an isomorphism from G_0 to G_1.

We set $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$. Note that $\Gamma_{\mathcal{G}} = I_f$ and $Y_{\mathcal{G}} = \{\mathrm{Orb}_{I_f}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}$. We show that $f \simeq f_{\mathcal{G}}$. Define $\sigma : Y \to Y_{\mathcal{G}}$ as follows: For any $y \in Y$, pick any $(x_0, x_1) \in f^{-1}(y)$ and set

$$\sigma(y) = \mathrm{Orb}_{I_f}(x_0, x_1).$$

We see that σ is well-defined. Indeed, the condition 3 ensures that for any $(x_0', x_1') \in f^{-1}(y)$, it holds that $\mathrm{Orb}_{I_f}(x_0', x_1') = f^{-1}(f(y)) = \mathrm{Orb}_{I_f}(x_0, x_1)$. We also see that σ is a bijection. We define $\sigma' : Y_{\mathcal{G}} \to Y$ as

$$\sigma'(\mathrm{Orb}_{I_f}(x_0, x_1)) = f(x_0, x_1)$$

for any $\mathrm{Orb}_{I_f}(x_0, x_1) \in Y_{\mathcal{G}}$. We see that σ' is well-defined. Indeed, if $\mathrm{Orb}_{I_f}(x_0, x_1) = \mathrm{Orb}_{I_f}(x_0', x_1')$, then there is $\pi \in I_f$ such that $\pi(x_0, x_1) = (x_0', x_1')$, which implies that $f(x_0', x_1') = f(\pi(x_0, x_1)) = f(x_0, x_1)$. Since σ' is clearly the inverse of σ, σ is a bijection. Furthermore, $\sigma(f(x_0, x_1)) = \mathrm{Orb}_{I_f}(x_0, x_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) = f_{\mathcal{G}}(x_0, x_1)$ for any $(x_0, x_1) \in X_0 \times X_1$. □

Example 4.6. The equality function $\mathsf{EQ}_N : [N] \times [N] \to \{0, 1\}$, defined as

$$\mathsf{EQ}_N(x_0, x_1) = \begin{cases} 1, & \text{if } x_0 = x_1, \\ 0, & \text{otherwise,} \end{cases}$$

is an ideal PSM function. Indeed, set $G_0 = G_1 = \mathcal{S}_N$ and set ψ as the identity function $\mathcal{I}$ from $\mathcal{S}_N$ to $\mathcal{S}_N$. Let $(x_0, x_1), (x_0', x_1') \in [N] \times [N]$ be such that $\mathsf{EQ}_N(x_0, x_1) = \mathsf{EQ}_N(x_0', x_1') =: y$. If $y = 0$, then $x_0 \neq x_1$ and $x_0' \neq x_1'$, and we have $\sigma(x_0, x_1) = (x_0', x_1')$ for $\sigma = (x_0\ x_0')(x_1\ x_1')$, which is a permutation that swaps two pairs (x_0, x_0') and (x_1, x_1'), respectively, and does not move any other elements. If $y = 1$, then let $a := x_0 = x_1$ and $a' := x_0' = x_1'$, and $b, b' \in [N]$ be two distinct elements other than a or a'. Then, $\sigma(x_0, x_1) = (x_0', x_1')$ for $\sigma = (a\ a')(b\ b')$. Thus, the fourth condition in Theorem 4.5 is satisfied by choosing $S = \{(\sigma, \sigma) : \sigma \in \mathcal{S}_N\}$.

The tuple $\mathcal{G} = (G_0, G_1, \psi)$ such that $f \simeq f_{\mathcal{G}}$ may not be uniquely determined by a function f. In other words, there may exist distinct $\mathcal{G} \neq \mathcal{G}' \in \Psi_{X_0,X_1}$ such that $f_{\mathcal{G}} \simeq f_{\mathcal{G}'}$. For example, while the equality function EQ_N is equivalent to $f_{\mathcal{G}}$ with $\mathcal{G} = (\mathcal{S}_N, \mathcal{S}_N, \mathcal{I})$, it is also equivalent to $f_{\mathcal{G}'}$ with $\mathcal{G}' = (A_N, A_N, \mathcal{I}|_{A_N})$. Here, A_N is the alternating group of degree N, which consists of all even permutations. Nevertheless, if $\mathcal{G}$ and $\mathcal{G}'$ are "conjugate," then they determine the unique function up to equivalence, as shown in the following proposition.

Proposition 4.7. *Let $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$. For $i \in \{0, 1\}$, let G_i' be a subgroup of $\mathcal{S}_{X_i}$ that is conjugate to G_i. Define $\psi' : G_0' \to G_1'$ as $\psi'(\tau_0^{-1} g_0 \tau_0) = \tau_1^{-1} \psi(g_0) \tau_1$ for any $g_0 \in G_0$. Then, ψ' is an isomorphism between G_0' and G_1', i.e., $\mathcal{G}' := (G_0', G_1', \psi') \in \Psi_{X_0,X_1}$. Furthermore, $f_{\mathcal{G}} \simeq f_{\mathcal{G}'}$.*

Proof. ψ' is an isomorphism since it is a composition of the following three isomorphisms:

- $G_0' \ni \tau_0^{-1} \pi_0 \tau_0 \mapsto \pi_0 \in G_0$
- $G_0 \ni \pi_0 \mapsto \psi(\pi_0) \in G_1$
- $G_1 \ni \pi_1 \mapsto \tau_1^{-1} \pi_1 \tau_1 \in G_1'$.

Thus, $\mathcal{G}' = (G'_0, G'_1, \psi') \in \Psi_{X_0,X_1}$.

It holds that $\tau^{-1}\Gamma_{\mathcal{G}}\tau = \Gamma_{\mathcal{G}'}$, where $\tau = (\tau_0, \tau_1)$. Indeed, for any $\pi = (\pi_0, \psi(\pi_0)) \in \Gamma_{\mathcal{G}}$, we have that

$$\tau^{-1}\pi\tau = (\tau_0^{-1}\pi_0\tau_0, \tau_1^{-1}\psi(\pi_0)\tau_1) = (\tau_0^{-1}\pi_0\tau_0, \psi'(\tau_0^{-1}\pi_0\tau_0)) \in \Gamma_{\mathcal{G}'}$$

and hence $\tau^{-1}\Gamma_{\mathcal{G}}\tau \subseteq \Gamma_{\mathcal{G}'}$. The converse inclusion is similar.

Next, we define $\sigma : \mathrm{Range}(f_{\mathcal{G}}) \to \mathrm{Range}(f_{\mathcal{G}'})$ by

$$\sigma(\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x)) = \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(\tau^{-1}(x)).$$

for any $x = (x_0, x_1) \in X_0 \times X_1$. σ is well-defined since if $x' = \pi(x)$ for some $\pi \in \Gamma_{\mathcal{G}}$, then $\tau^{-1}(x') = \tau^{-1}\pi(x) = (\tau^{-1}\pi\tau)(\tau^{-1}(x)) \in \mathrm{Orb}_{\tau^{-1}\Gamma_{\mathcal{G}}\tau}(\tau^{-1}(x)) = \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(\tau^{-1}(x))$. σ is injective since for any $x, x' \in X_0 \times X_1$,

$$\begin{aligned}
\mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(\tau^{-1}(x)) = \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(\tau^{-1}(x')) &\Leftrightarrow \exists \pi' \in \Gamma_{\mathcal{G}'} : \pi'\tau^{-1}(x) = \tau^{-1}(x') \\
&\Leftrightarrow \exists \pi \in \Gamma_{\mathcal{G}} : (\tau^{-1}\pi\tau)(\tau^{-1}(x)) = \tau^{-1}(x') \\
&\Leftrightarrow \exists \pi \in \Gamma_{\mathcal{G}} : \pi(x) = x' \\
&\Leftrightarrow \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(x) = \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(x').
\end{aligned}$$

σ is surjective (and hence a bijection) since $\mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(x) = \sigma(\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(\tau(x)))$ for any x.

Then, $f_{\mathcal{G}} \simeq f_{\mathcal{G}'}$ follows from $\sigma(f_{\mathcal{G}}(\tau(x))) = \sigma(\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(\tau(x))) = \mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(x) = f_{\mathcal{G}'}(x)$. □

Remark 4.8. We can show that Theorem 4.5 holds even if we relax the privacy requirement of a PSM scheme for f. More precisely, the conditions in Theorem 4.5 are also equivalent to the following condition:

1'. f admits a weakly private ideal PSM scheme. That is, there is a PSM scheme $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ for f satisfying the perfect correctness condition (1) and the weak privacy condition (3).

In other words, the set of functions admitting weakly private ideal PSM schemes is identical with those admitting perfectly private ideal PSM schemes. A sketch of the proof is as follows. First, Lemma 4.3 holds even if Π is a weakly private PSM scheme since the perfect correctness is only used in the proof. In addition, a statement that is analogous to Lemma 4.4 holds true for weakly private schemes. Indeed, in the proof of Lemma 4.4, if Π is weakly private, then so is Π', since the statistical distance does not increase when deterministic functions μ_i are applied. Now, we can prove an implication **1'** $\Rightarrow$ **4**, which is sufficient as **1** $\Rightarrow$ **1'** is trivial. The only argument where we used the perfect privacy in the proof of the implication **1** $\Rightarrow$ **4** was showing the existence of $r, r' \in R$ such that $(\pi_r^0(x_0), \pi_r^1(x_1)) = (\pi_{r'}^0(x'_0), \pi_{r'}^1(x'_1))$ in its third paragraph. In fact, the weak privacy is sufficient for this since the existence of r and r' follows from the fact that the supports of the message distributions $\mathcal{D}_{(x_0,x_1)}$, $\mathcal{D}_{(x'_0,x'_1)}$ have a non-empty intersection. The remaining argument is the same.

4.3 Basic Properties

Based on the characterization, we show some basic properties of ideal PSM functions.

Proposition 4.9. *Let $f : X_0 \times X_1 \to Y$ be a non-degenerate connected function that admits ideal PSM. Then, for any $x_0, x'_0 \in X_0$, the row vectors $T_f[x_0, *], T_f[x'_0, *] \in Y^{|X_1|}$ are identical as multisets (that is, one is a permutation of the other). Similarly, for any $x_1, x'_1 \in X_1$, the column vectors $T_f[*, x_1], T_f[*, x'_1] \in Y^{|X_0|}$ are identical as multisets.*

Proof. we may assume that $f = f_{\mathcal{G}}$ for some $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$. Let $x_0, x'_0 \in X_0$. First, we consider the case where there is an edge between x_0 and x'_0 in the graph $\mathfrak{G}_{f,0}$. Then, $f(\{x_0\} \times X_1) \cap f(\{x'_0\} \times X_1) \neq \emptyset$ and hence there exist $a_1, a'_1 \in X_1$ such that $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, a_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x'_0, a'_1)$. In particular, there exists $(\pi_0, \pi_1) \in \Gamma_{\mathcal{G}}$ such that $(x'_0, a'_1) = (\pi_0(x_0), \pi_1(a_1))$. Thus, for any $x_1 \in X_1$, it holds that $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(\pi_0(x_0), \pi_1(x_1)) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x'_0, \pi_1(x_1))$, implying that $T_f[x_0, *]$ and $T_f[x'_0, *]$ are identical as multisets. For the general case, since $\mathfrak{G}_{f,0}$ is a connected graph, there are a sequence of edges between x_0 and x_1. By applying the above argument to each edge, we conclude that $T_f[x_0, *]$ and $T_f[x'_0, *]$ are identical as multisets. □

We can express a boolean function $f : X_0 \times X_1 \to \{0,1\}$ as a bipartite graph B_f whose vertex set consists of $X_0 \cup X_1$ and whose edge set is $E_f = \{(x_0, x_1) \in X_0 \times X_1 : f(x_0, x_1) = 1\}$. We say that a graph is *edge-transitive* if for any pair of edges e, e' in the graph, there exists a graph automorphism that maps e to e'. We can show that bipartite graphs induced by ideal PSM functions are edge-transitive.

Proposition 4.10. *Let $f : X_0 \times X_1 \to \{0,1\}$ be a non-degenerate boolean function. If f is an ideal PSM function, then the bipartite graphs B_f and $B_{\overline{f}}$ associated with f and $\overline{f}$ are both edge-transitive. Furthermore, if $|X_0| \neq |X_1|$, then the converse is also true.*

Proof. Let $e = (x_0, x_1), e' = (x_0', x_1')$ be any pair of edges in B_f. By definition, $f(x_0, x_1) = f(x_0', x_1') = 1$, i.e., $(x_0, x_0), (x_0', x_1') \in f^{-1}(1)$. Thus, Theorem 4.5 ensures that there exists $\pi = (\pi_0, \pi_1) \in I_f$ such that $(\pi_0(x_0), \pi_1(x_1)) = (x_0', x_1')$. It suffices to see that π is a graph automorphism of B_f. Let $e'' = (x_0'', x_1'') \in X_0 \times X_1$. Note that the orbits of e'' and $\pi(e'')$ are identical, i.e., $\mathrm{Orb}_{I_f}(x_0'', x_1'') = \mathrm{Orb}_{I_f}(\pi_0(x_0''), \pi_1(x_1''))$. Theorem 4.5 ensures that the orbit is equal to $f^{-1}(0)$ or $f^{-1}(1)$. Thus, e'' is an edge in E_f if and only if $\pi(e'')$ is an edge in E_f. We can show that $B_{\overline{f}}$ is edge-transitive in a similar way.

To see the converse, let $S = \mathrm{Aut}(B_f) \leq \mathcal{S}_{X_0 \cup X_1}$ be the automorphism group of B_f. Note that $S = \mathrm{Aut}(B_{\overline{f}})$ since the set of edges in $B_{\overline{f}}$ is the complement of the set of edges in B_f. In addition, if $\sigma \in S$, then $\sigma(X_0) = X_0$ and $\sigma(X_1) = X_1$. This is because otherwise, $\sigma(X_0) = X_1$ and $\sigma(X_1) = X_0$ [42], which contradicts the assumption that $|X_0| \neq |X_1|$. Thus, we may assume that $S \subseteq \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ under the correspondence $\sigma \mapsto (\sigma|_{X_0}, \sigma|_{X_1})$.

We have $S \subseteq I_f$ since for any $\sigma \in S$ and $e = (x_0, x_1) \in X_0 \times X_1$,

$$f(\sigma|_{X_0}(x_0), \sigma|_{X_1}(x_1)) = 1 \ \Leftrightarrow \ \sigma(e) \in E_f \ \Leftrightarrow \ e \in E_f \ \Leftrightarrow \ f(x_0, x_1) = 1.$$

Furthermore, let $e = (x_0, x_1), e' = (x_0', x_1') \in X_0 \times X_1$ be such that $f(x_0, x_1) = f(x_0', x_1')$. If $f(x_0, x_1) = f(x_0', x_1') = 1$, then $e, e' \in E_f$. Since B_f is edge-transitive, there is an automorphism $\sigma = (\sigma_0, \sigma_1) \in S$ such that $e' = \sigma(e)$, that is, $(x_0', x_1') = (\sigma_0(x_0), \sigma_1(x_1))$. If $f(x_0, x_1) = f(x_0', x_1') = 0$, then $e, e' \in E_{\overline{f}}$. Since $B_{\overline{f}}$ is edge-transitive, there is an automorphism $\tau = (\tau_0, \tau_1) \in \mathrm{Aut}(B_{\overline{f}})$ such that $e' = \tau(e)$. Since $\mathrm{Aut}(B_{\overline{f}}) = S$, we have $(x_0', x_1') = (\tau_0(x_0), \tau_1(x_1))$ and $\tau \in S$. Thus, $\mathrm{Orb}_S(x_0, x_1) = f^{-1}(f(x_0, x_1))$ for any (x_0, x_1). From Theorem 4.5, f admits ideal PSM. $\square$

Finally, we show some properties that will be used to count the total number of ideal PSM functions.

Proposition 4.11. *Let $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0, X_1}$. If $f_{\mathcal{G}}$ is a connected function, then both G_0 and G_1 are transitive groups.*

Proof. We simply write $f = f_{\mathcal{G}}$. We consider the case of $i = 0$ (the case of $i = 1$ is similar). Let $x_0, x_0' \in X_0$. As shown in Section 3.2, the graph $\mathfrak{G}_{f,0}$ is connected. Thus, there exists a sequence of vertices $a_0 = x_0, a_1, \ldots, a_{m-1}, a_m = x_0' \in X_0$ such that there is an edge between each pair (a_j, a_{j+1}). This implies that for each j, $f_{\mathcal{G}}(\{a_j\} \times X_1) \cap f_{\mathcal{G}}(\{a_{j+1}\} \times X_1) \neq \emptyset$ and hence there are $x_1, x_1' \in X_1$ such that $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(a_j, x_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(a_{j+1}, x_1')$. In particular, there exists $\pi_0^j \in G_0$ such that $a_{j+1} = \pi_0^j(a_j)$. Therefore, we have $x_0' = \pi_{m-1} \circ \cdots \circ \pi_0(x_0)$, implying that G_0 is transitive. $\square$

Proposition 4.12. *Assume that $|X_0|$ and $|X_1|$ are primes. Let $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0, X_1}$. If $f_{\mathcal{G}}$ is a connected function, then $|X_0| = |X_1|$.*

Proof. Let $N_0 = |X_0|$ and $N_1 = |X_1|$. Proposition 4.11 ensures that G_0 and G_1 are transitive permutation groups of degree N_0 and N_1, respectively. Then, it holds that $|G_1| = |X_1| \times |\mathrm{Stab}_{G_1}(x_1)|$ for any $x_1 \in X_1$, where $\mathrm{Stab}_{G_1}(x_1) = \{\pi_1 \in G_1 : \pi_1(x_1) = x_1\}$. In particular, $|G_1|$ is a multiple of N_1. Since G_0 and G_1 are isomorphic, $|G_0| = |G_1|$ is also a multiple of N_1 and hence $N_1 \leq N_0$ because $N_0!$ can be divided by $|G_0|$ and N_1 is a prime. By the same argument, we can prove that $N_0 \leq N_1$. Therefore, we have $N_0 = N_1$. $\square$

5 Enumeration of Ideal PSM Functions

In this section, we give asymptotic upper bounds on the total number of non-degenerate connected functions $f : X_0 \times X_1 \to Y$ that admit ideal PSM. In addition, we provide a complete list of such functions for small domains by a brute-force search. Our approach for enumeration is based on Theorem 4.5, which ensures that it is sufficient to enumerate all possible tuples $(G_0, G_1, \psi) \in \Psi_{X_0,X_1}$ where $G_0 \leq \mathcal{S}_{X_0}$, $G_1 \leq \mathcal{S}_{X_1}$, and ψ is an isomorphism from G_0 to G_1.

Terminology from Group Theory. A group G is called *solvable* if there exists a sequence $(G_0, \ldots, G_n)$ of subgroups such that $G_0 = \{1_G\}$ (i.e., the trivial group), $G_n = G$, G_i is a normal subgroup of G_{i+1}, and the quotient group G_{i+1}/G_i is abelian. A group G is called *simple* if G does not have any normal subgroups other than the trivial group or G itself. A group G is called *almost simple* if there exists a non-abelian simple group S such that $S \leq G \leq \mathrm{Aut}(S)$, where $\mathrm{Aut}(S)$ denotes the group of all automorphisms of S.

Let X be a non-empty set and let G be a transitive subgroup of $\mathcal{S}_X$. A partition $\mathcal{B} = (B_i)_{i \in I}$ of X is called a *block system of* G if for any $g \in G$ and $B_i \in \mathcal{B}$, a set $\{g(x) : x \in B_i\}$ is also an element of $\mathcal{B}$, that is, there exist $j \in I$ such that $\{g(x) : x \in B_i\} = B_j$. A block system $\mathcal{B}$ is called trivial if $\mathcal{B} = \{X\}$ or $\mathcal{B} = \{\{x\} : x \in X\}$. A group G is called *primitive* if G has no non-trivial block systems. A (not necessarily transitive) subgroup $G \leq \mathcal{S}_X$ is called *orbit-minimal* if for any proper subgroup $H \lneq G$, it holds that $|\{\mathrm{Orb}_H(x) : x \in X\}| > |\{\mathrm{Orb}_G(x) : x \in X\}|$.

5.1 Asymptotic Upper Bounds

We define C_{N_0,N_1} as the total number of non-degenerate connected functions f, up to the equivalence relation $\simeq$ (defined in Section 3.2), such that

- The domain of f is $X_0 \times X_1$ with $|X_0| = N_0$ and $|X_1| = N_1$.
- f admits ideal PSM.

From Theorem 4.5, a non-degenerate ideal PSM function f is equivalent to $f_{\mathcal{G}}$ for some $\mathcal{G} \in \Psi_{X_0,X_1}$. Thus, we can upper bound C_{N_0,N_1} by enumerating all $f_{\mathcal{G}}$'s up to equivalence.

Furthermore, it suffices to enumerate all orbit-minimal subgroups H of $\mathcal{S}_{X_0 \times X_1}$ such that $H \leq \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$. Indeed, given $f_{\mathcal{G}}$, let $H_{\mathcal{G}}$ be a minimal one among subgroups of $\Gamma_{\mathcal{G}}$ such that

$$\{\mathrm{Orb}_{H_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\} = \{\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}.$$

Then, $H_{\mathcal{G}}$ is an orbit-minimal subgroup of $\mathcal{S}_{X_0 \times X_1}$ since otherwise, there is a proper subgroup $H' \lneq H_{\mathcal{G}}$ such that

$$\begin{aligned} |\{\mathrm{Orb}_{H'}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}| &= |\{\mathrm{Orb}_{H_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}| \\ &= |\{\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\}|, \end{aligned}$$

which contradicts the choice of $H_{\mathcal{G}}$. Furthermore, if $f_{\mathcal{G}}$ and $f_{\mathcal{G}'}$ lead to the same subgroup $H = H_{\mathcal{G}} = H_{\mathcal{G}'}$, then it holds that

$$\begin{aligned} \{\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\} &= \{\mathrm{Orb}_{H}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\} \\ &= \{\mathrm{Orb}_{\Gamma_{\mathcal{G}'}}(x_0, x_1) : (x_0, x_1) \in X_0 \times X_1\} \end{aligned}$$

and hence $f_{\mathcal{G}} \simeq f_{\mathcal{G}'}$. Therefore, the total number of non-equivalent $f_{\mathcal{G}}$'s and hence C_{N_0,N_1} are upper bounded by the total number of orbit-minimal subgroups H of $\mathcal{S}_{X_0 \times X_1}$ such that $H \leq \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$.

As for orbit-minimal permutation groups, we have the following proposition [45, Theorem 1.5].

Fact 5.1. *Any orbit-minimal subgroup G of $\mathcal{S}_X$ can be generated by $\log(|X|-k+1)$ elements where k denotes the number of orbits of G, i.e., $k := |\{\mathrm{Orb}_G(x) : x \in X\}|$.*

Thus, any orbit-minimal subgroup H with $H \le \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ is generated by a set S of at most $\log(N_0N_1 - k+1) \le \log(N_0N_1)$ elements. Furthermore, since each element of S is expressed as (π_0, π_1) for some $\pi_0 \in \mathcal{S}_{X_0}$ and $\pi_1 \in \mathcal{S}_{X_1}$, the total number of such H's is upper bounded by

$$\binom{|\mathcal{S}_{X_0}| \cdot |\mathcal{S}_{X_1}|}{\log(N_0N_1)}.$$

Theorem 5.2. *The total number of non-degenerate ideal PSM functions $f : X_0 \times X_1 \to Y$ with $|X_0| = N_0$ and $|X_1| = N_1$, up to equivalence, is at most*

$$\binom{N_0! \cdot N_1!}{\log(N_0N_1)} = 2^{O((N_0 \log N_0 + N_1 \log N_1) \log N_0N_1)}$$

Note that we did not use the assumption that f is connected in the above argument. Thus, Theorem 5.2 gives an upper bound on the number of possibly disconnected ideal PSM functions.

Special Case. If the message spaces of both parties are of prime size, i.e., N_0 and N_1 are primes, then we can obtain a better upper bound for C_{N_0,N_1}.

Let $\mathscr{T}_N$ be the set of all transitive subgroups of $\mathcal{S}_N$, up to conjugacy. From Propositions 4.7 and 4.11, C_{N_0,N_1} is upper bounded by the total number of non-equivalent $f_\mathcal{G}$'s for $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{X_0,X_1}$ such that $G_0 \in \mathscr{T}_{N_0}$ and $G_1 \in \mathscr{T}_{N_1}$. From Proposition 4.12, we may also assume that $p := N_0 = N_1$ and $X_0 = X_1 = [p]$.

Let Ψ'_{X_0,X_1} be the set consisting of all $\mathcal{G}$'s except for those such that $f_\mathcal{G}$ is equivalent to the equality function EQ_p. Formally, we define

$$\Psi'_{X_0,X_1} = \Psi_{X_0,X_1} \setminus \{\mathcal{G} : f_\mathcal{G} \simeq \mathsf{EQ}_p\}.$$

We also define

$$\mathscr{G} = \{(G_0, G_1) \in \mathscr{T}_{N_0} \times \mathscr{T}_{N_1} : (G_0, G_1, \psi) \in \Psi'_{X_0,X_1} \text{ for some } \psi\}.$$

Then, since it is sufficient to consider orbit-minimal subgroups as above, C_{N_0,N_1} can be upper bounded as

$$\begin{aligned} C_{N_0,N_1} &\le |\{\mathsf{EQ}_p\}| + |\{H_\mathcal{G} : \mathcal{G} \in \Psi'_{X_0,X_1}\}| \\ &\le 1 + \sum_{(G_0,G_1)\in\mathscr{G}} |\{H \le \mathcal{S}_{X_0 \times X_1} : H \text{ is orbit-minimal and } H \le G_0 \times G_1\}|. \end{aligned}$$

Since each such H is generated by at most $\log(N_0N_1) = 2\log p$ elements of $G_0 \times G_1$, we have

$$C_{N_0,N_1} \le 1 + \sum_{(G_0,G_1)\in\mathscr{G}} \binom{|G_0| \cdot |G_1|}{\log(N_0N_1)} \le 1 + \sum_{(G_0,G_1)\in\mathscr{G}} |G_0|^{2\log p} |G_1|^{2\log p}$$

We will prove the following fact.

Proposition 5.3. *It holds that $|\mathscr{T}_{N_0}| = |\mathscr{T}_{N_1}| = |\mathscr{T}_p| = 2^{O(\log p)}$ and that $|G_0| = |G_1| = 2^{O(\log^2 p)}$ for any $(G_0, G_1) \in \mathscr{G}$.*

Then, we obtain an upper bound on C_{N_0,N_1} as

$$C_{N_0,N_1} \le 1 + |\mathscr{T}_{N_0}| \cdot |\mathscr{T}_{N_1}| \cdot (2^{O(\log^2 p)})^{2\log p} \cdot (2^{O(\log^2 p)})^{2\log p} \le 2^{O(\log^3 p)}.$$

Hence, we obtain the following theorem.

Theorem 5.4. *Let N_0 and N_1 be primes and $f : X_0 \times X_1 \to Y$ be a non-degenerate, connected, ideal PSM function with $|X_0| = N_0$ and $|X_1| = N_1$. Then, $N_0 = N_1$. Furthermore, the total number C_{N_0,N_1} of such functions is at most*

$$C_{N_0,N_1} \le 2^{O(\log^3 p)},$$

where $p := N_0 = N_1$.

In the following, we provide the proof of Proposition 5.3.

(1) Upper bound on $|\mathscr{T}_p|$. Any transitive permutation groups of degree p are primitive. The number of conjugacy classes of primitive permutation groups of degree p is upper bounded by p^c, where c is a constant [46, Theorem 1]. Thus, we have $|\mathscr{T}_{N_0}| = |\mathscr{T}_{N_1}| = |\mathscr{T}_p| = 2^{O(\log p)}$.

(2) Upper bound on $|G_0|$ and $|G_1|$ for $(G_0, G_1) \in \mathscr{G}$. We consider an upper bound on $|G_0|$ for $G_0 \in \mathscr{T}_p$, which also applies to $|G_1|$ as $|G_0| = |G_1|$. The following fact is known [30, Theorem 2.37]:

Fact 5.5. *For any $G \in \mathscr{T}_p$, G is solvable or almost simple.*

If G_0 is solvable, we can use a known upper bound on the order of any primitive solvable group [45, Theorem 2.7]:

Fact 5.6. *Let G be a primitive solvable subgroup of $\mathcal{S}_p$. Then $|G| \le 2^{O(\log p)}$.*

Assume that G_0 is almost simple. By definition, there exists a non-abelian simple group S such that $S \le G_0 \le \mathrm{Aut}(S)$. The following fact is known [30, Corollary 2.39]:

Fact 5.7. *If $G \in \mathscr{T}_p$ is almost simple and S is a non-abelian simple group such that $S \le G \le \mathrm{Aut}(S)$, then S is one of the following groups:*

(A) $S = A_p$, where A_p is an alternating group of degree p.

(B) $S = \mathrm{PSL}(n, q)$ with $p = (q^n - 1)/(q - 1)$ and n is prime.

(C) $S = \mathrm{PSL}(2, 11)$ with $p = 11$.

(D) $S = M_p$ with $p = 11$ or $p = 23$, where M_p is a Mathieu groups.

We may ignore the cases (C) and (D) since $|S| = O(1)$ and hence $|G_0| \le |\mathrm{Aut}(S)| = O(1)$.

First, we consider the case (B). In this case, $|G_0|$ is upper bounded by $\mathrm{Aut}(\mathrm{PSL}(n, q))$, where $p = N_0 = (q^n - 1)/(q - 1)$ and n is a prime. In particular, we have $n = O(\log p)$. Since $|\mathrm{Aut}(\mathrm{PSL}(n, q))|$ is upper bounded by $|\mathrm{PSL}(n, q)| \cdot (p-1)$ [30, Theorem 2.47], we have $|G_0| \le |\mathrm{PSL}(n, q)| \cdot (p-1) \le q^{O(n^2)} \cdot p = 2^{O(\log^2 p)}$.

Next, we show that the case (A) does not happen if G_0 is such that $(G_0, G_1) \in \mathscr{G}$. Assume to the contrary that $(G_0, G_1) \in \mathscr{G}$ and $S = A_p$. Then, G_0 is either A_p or $\mathcal{S}_p$. Also, there is an isomorphism $\psi : G_0 \to G_1$ such that $\mathcal{G} = (G_0, G_1, \psi) \in \Psi'_{X_0, X_1}$. We will prove that $f = f_{\mathcal{G}}$ is degenerate or equivalent to the equality function, which leads to contradiction.

Recall that for $x_1 \in X_1$, we denote by $T_f[*, x_1]$ the column vector of the truth-table T_f corresponding to x_1. Also, $\pi \in \mathcal{S}_{X_0}$ naturally acts on the set of the column vectors $T_f[*, x_1]$'s. By the definition of $f_{\mathcal{G}}$, we have $\pi_0(T_f[*, x_1]) = T_f[*, (\psi(\pi_0))^{-1}(x_1)]$ for any $\pi_0 \in G_0$ and $x_1 \in X_1$. This implies that G_0 can be regarded as a transitive permutation group acting on the set $\{T_f[*, 0], \ldots, T_f[*, p-1]\}$ and therefore we have $|G_0| = pm$ where $m = |\{\pi_0 \in G_0 : \pi_0(v_0) = v_0\}|$.

For $y \in Y = \mathrm{Range}(f_{\mathcal{G}})$, let k_y denote the number of occurrences of y in v_0. Then, we have

$$m = |\{\pi_0 : \pi_0 \in G_0, \pi_0(v_0) = v_0\}| \le |\{\pi_0 : \pi_0 \in \mathcal{S}_{X_0}, \pi_0(v_0) = v_0\}| = \prod_{y \in Y} (k_y)!.$$

Let $\ell = |Y|$ and assume that $Y = [\ell]$ (without loss of generality). We may also assume that

$$1 \le k_0 \le k_1 \le \cdots \le k_{\ell-1}. \tag{6}$$

Note that $\sum_{i \in [\ell]} k_i = p$. For any sequence $(k_0, \ldots, k_{\ell-1})$ satisfying (6), we let $M(k_0, \ldots, k_{\ell-1})$ denote $\prod_{i \in [\ell]} (k_i)!$. If $k_{\ell-1} = p$ (and hence $\ell = 1$), then $\pi_0(v_0) = v_0$ for any $\pi_0 \in S_{X_0}$ and therefore $f_{\mathcal{G}}$ is degenerate. If $k_{\ell-1} = p - 1$ (and hence $\ell = 2$ and $k_0 = 1$), then $f_{\mathcal{G}}$ is equivalent to the equality function. Therefore, if $f_{\mathcal{G}}$ is non-degenerate and is not equivalent to the equality function, then $k_{\ell-1} \le p - 2$. However, we will then prove that

$$M(k_0, \ldots, k_{\ell-1}) \le \frac{(p-1)!}{2} \tag{7}$$

for any $p \geq 5$ and $(k_0, \dots, k_{\ell-1})$ with $k_{\ell-1} \leq p-2$. This implies $|G_0| = pm < p!/2$, which contradicts that $G_0 = A_p$ or $G_0 = \mathcal{S}_p$.

To see (7), we observe that $M(k_0, \dots, k_{\ell-1}) \leq M(k_0 - 1, k_1, \dots, k_{\ell-2}, k_{\ell-1} + 1)$. If $\ell = 2$, using this inequality recursively, we have $M(k_0, k_1) \leq M(2, p-2) = (p-2)! \cdot 2 \leq (p-1)!/2$ for $p \geq 5$. If $\ell > 2$, then we have $M(k_0, \dots, k_{\ell-1}) \leq M(1, k_1, \dots, k_{\ell-2}, k_{\ell-1} + k_0 - 1)$. Hence, repeating this procedure, we obtain $M(k_0, \dots, k_{\ell-1}) \leq M(1, \dots, 1, p - \ell + 1) \leq (p-2)! \leq (p-1)!/2$ for $p \geq 3$. This completes the proof of Proposition 5.3.

Fraction of Ideal PSM Functions. As a demonstration, we provide asymptotic upper bounds on the fraction of ideal PSM functions among the set of all non-degenerate functions in the boolean case. Note that the connectivity assumption does not affect the asymptotic bounds in the boolean case since non-degenerate disconnected boolean functions are equivalent to those whose truth-tables are (1 0) or its transpose. Thus, the asymptotic bounds in Theorems 5.2 and 5.4 can apply.

Let D_N denote the total number of non-degenerate functions $f : [N] \times [N] \to \{0, 1\}$ up to equivalence. First, if we do not exclude equivalent ones, the number of non-degenerate functions is lower bounded by

$$\begin{aligned} 2^{N^2} - (2^{N^2} - \prod_{i=0}^{N-1} (2^N - i)) - (2^{N^2} - \prod_{i=0}^{N-1} (2^N - i)) &= 2^{N^2}(-1 + 2\prod_{i=0}^{N-1} (1 - i/2^N)) \\ &\geq 2^{N^2}(-1 + 2(1 - N/2^N)^N) \\ &= 2^{N^2}(1 - o(1)). \end{aligned}$$

Note that $2^{N^2} - \prod_{i=0}^{N-1}(2^N - i)$ is the number of functions whose truth-tables contain identical rows. (The number of functions that contain identical columns is the same.) We can thus bound D_N from below as

$$D_N \geq \frac{2^{N^2}(1 - o(1))}{2(N!)^2} = 2^{N^2 - O(N \log N)},$$

where the factor of 2 in the denominator comes from $f \simeq \overline{f}$. Therefore, the fraction of non-degenerate ideal PSM functions is at most

$$\frac{C_{N,N}}{D_N} \leq \frac{2^{O(N \log^2 N)}}{2^{N^2 - O(N \log N)}} = \frac{2^{O(N \log^2 N)}}{2^{N^2}} = 2^{-N^2 + O(N \log^2 N)}.$$

Furthermore, if N is a prime, then the upper bound can be much smaller:

$$\frac{C_{N,N}}{D_N} \leq \frac{2^{O(\log^3 N)}}{2^{N^2 - O(N \log N)}} = 2^{-N^2 + O(N \log N)}.$$

These bounds demonstrate that the fraction of ideal PSM functions is exponentially small compared to the set of all non-degenerate functions up to equivalence.

5.2 Complete Lists of Small Ideal PSM Functions

Finally, we provide a complete list of non-degenerate connected functions $f : X_0 \times X_1 \to Y$ that admit ideal PSM when $|X_0|$ and $|X_1|$ are small. From Proposition 4.7, it is sufficient to enumerate $G_0 \leq \mathcal{S}_{X_0}$ and $G_1 \leq \mathcal{S}_{X_1}$ up to conjugacy, and isomorphism $\psi \colon G_0 \to G_1$. For small domains, we do this by a brute-force search using an open software package called GAP. Since we focus on connected functions, we enumerate such (G_0, G_1, ψ) assuming that G_0 and G_1 are transitive. GAP has a list of all transitive permutation groups (up to conjugacy) of degree at most 30. GAP has a function that takes G_0 and G_1 as input and computes an isomorphism from G_0 to G_1 (if exists). GAP also has a function that takes G_0 as input and computes all automorphism of G_0. Using these functions, we can enumerate Ψ_{X_0,X_1} with $\max\{|X_0|, |X_1|\} \leq 15$ except for the case of $|X_0| = |X_1| = 12$. We were unable to deal with larger cases or $|X_0| = |X_1| = 12$ due to limited computational resources. There are possibilities that $f_{\mathcal{G}}$ is degenerate for some $\mathcal{G} \in \Psi_{X_0,X_1}$ or that $f_{\mathcal{G}} \simeq f_{\mathcal{G}'}$ for some $\mathcal{G} \neq \mathcal{G}' \in \Psi_{X_0,X_1}$ (even if they are not conjugate). We manually eliminate such degenerate functions and duplicates. The resulting list and our source codes are available in [1] and the exact values of C_{N_0,N_1} are provided in Table 1. It can be observed that the exact values of C_{N_0,N_1} seem to be much smaller than our asymptotic upper bounds. We leave the question of the tightness of our upper bounds on C_{N_0,N_1} to future work.

More Efficient Enumeration for Boolean Functions. We show a more efficient method to enumerate *boolean* functions $f : X_0 \times X_1 \to \{0,1\}$ that admit ideal PSM. As a result, we can enumerate such functions with $\max\{|X_0|, |X_1|\} \leq 23$. Without loss of generality, we may assume that $X_0 = [n]$, $X_1 = [n']$, and $n' \leq n$. For a transitive group $G_0 \in \mathscr{T}_n$, we define a graph (V_{G_0}, E_{G_0}) as follows:

$$V_{G_0} = \{0,1\}^n \text{ and } E_{G_0} = \{(v, v') \in (\{0,1\}^n)^2 : \exists \pi_0 \in G_0, v' = \pi_0(v)\}.$$

Assume that $f = f_{\mathcal{G}}$ for some $\mathcal{G} = (G_0, G_1, \psi) \in \Psi_{[n],[n']}$, where $G_1 \leq \mathcal{S}_{n'}$ and $\psi : G_0 \to G_1$ is an isomorphism. Here, we interpret that each output $y \in \{0,1\}$ of f corresponds to the orbit $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1)$ with $f(x_0, x_1) = y$. If the truth-table T_f contains $v \in \{0,1\}^n$ as a column vector, then T_f also contains $\pi_0(v)$ as a column vector for any $\pi_0 \in G_0$. Indeed, if $v = T_f[*, x_1]$ for some $x_1 \in X_1$, then $\mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, x_1) = v_{x_0}$ for any $x_0 \in X_0$. Since $(\pi_0, \pi_1) = (\pi_0, \psi(\pi_0)) \in \Gamma_{\mathcal{G}}$, we have $v_{\pi_0(x_0)} = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(\pi_0(x_0), x_1) = \mathrm{Orb}_{\Gamma_{\mathcal{G}}}(x_0, \pi_1^{-1}(x_1))$ for any $x_0 \in X_0$, which implies that $\pi_0(v)$ appears in $T_f[*, \pi_1^{-1}(x_1)]$. Conversely, since we assume that f is connected, Proposition 4.9 ensures that for any two column vectors v, v' in T_f, there exists $\pi_0 \in G_0$ such that $v' = \pi_0(v)$. Therefore, such f corresponds to a connected component of the graph (V_{G_0}, E_{G_0}) whose size is n'. Hence, we can enumerate ideal PSM functions $f : [n] \times [n'] \to \{0,1\}$ with $n' \leq n$ as follows: For every transitive subgroup $G_0 \in \mathscr{T}_n$ up to conjugacy,

1. Construct the graph (V_{G_0}, E_{G_0}).
2. Compute the set $\mathcal{C}_{G_0}$ of all connected components of (V_{G_0}, E_{G_0}) whose size is at most n.

Finally, we collect all such components into $\mathcal{C}$, that is, $\mathcal{C} = \bigcup_{G_0} \mathcal{C}_{G_0}$. $\mathcal{C}$ covers the set of all ideal PSM functions. Since $\mathcal{C}$ may contain degenerate functions or duplications, we manually eliminate them. The resulting list is available in [1] and the exact values of C_{N_0,N_1} are provided in Table 2.

6 Infinite Families of Ideal PSM Functions

In this section, we provide several infinite families of ideal PSM functions.

6.1 Group Product

The sum function $f : G \times G \to G$ over an abelian group G, defined as $f(x, y) = x + y$, admits ideal PSM: one party with input x computes a message $x + r$ and the other party with input y computes a message $y - r$.

More generally, let G be a possibly non-abelian group. Let $m \in \mathbb{N}$, (S_0, S_1) be a partition of $[m]$, and $s_i = |S_i|$ for $i \in \{0,1\}$. We define a function $f : G^{s_0} \times G^{s_1} \to G$ as

$$f((x_i)_{i \in S_0}, (y_i)_{i \in S_1}) = \prod_{i \in [m]} z_i,$$

where we set $z_i = x_i$ for $i \in S_0$ and $z_i = y_i$ for $i \in S_1$. Then, f admits the following ideal PSM scheme [29]:

1. The shared randomness is uniformly random group elements $(r_i)_{i \in [m+1]}$ such that r_0 and r_m are the identity element of G.
2. One party with input $(x_i)_{i \in S_0} \in G^{s_0}$ sends $(r_i^{-1} x_i r_{i+1})_{i \in S_0} \in G^{s_0}$. The other party with input $(y_i)_{i \in S_1} \in G^{s_1}$ sends $(r_i^{-1} y_i r_{i+1})_{i \in S_1} \in G^{s_1}$.
3. Upon receiving two messages $(w_i)_{i \in S_0}$ and $(w_i)_{i \in S_1}$, the output is obtained by $\prod_{i \in [m]} w_i$.

6.2 Private Set Intersection Cardinality

Let U be a set of size n and $d_0, d_1 \in \mathbb{N}$ be such that $d_0 \le n$ and $d_1 \le n$. Set $d = \min\{d_0, d_1\}$. Define a function $\mathsf{PSIC}_{n,(d_0,d_1)} : \binom{U}{d_0} \times \binom{U}{d_1} \to \{0, 1, \dots, d\}$ as

$$\mathsf{PSIC}_{n,(d_0,d_1)}(A_0, A_1) = |A_0 \cap A_1| \tag{8}$$

for all $(A_0, A_1) \in \binom{U}{d_0} \times \binom{U}{d_1}$. We show that $\mathsf{PSIC}_{n,(d_0,d_1)}$ admits ideal PSM based on Theorem 4.5. Let $(A_0, A_1), (A'_0, A'_1) \in \binom{U}{d_0} \times \binom{U}{d_1}$ be two inputs such that $|A_0 \cap A_1| = |A'_0 \cap A'_1|$. Observe that both $A_0 \cup A_1$ and $A'_0 \cup A'_1$ admit partitions whose corresponding blocks have the same sizes, that is,

$$A_0 \cup A_1 = (A_0 \setminus A_1) \cup (A_0 \cap A_1) \cup (A_1 \setminus A_0),$$
$$A'_0 \cup A'_1 = (A'_0 \setminus A'_1) \cup (A'_0 \cap A'_1) \cup (A'_1 \setminus A'_0).$$

Thus, there exists a permutation σ over U such that $\sigma(A_0) = A'_0$ and $\sigma(A_1) = A'_1$. Furthermore, since σ is a permutation, $|\sigma(X) \cap \sigma(Y)| = |\sigma(X \cap Y)| = |X \cap Y|$ for any $(X, Y) \in \binom{U}{d_0} \times \binom{U}{d_1}$. Thus, the fourth condition in Theorem 4.5 is satisfied.

If we set $d_0 = d_1 = 1$, then $\mathsf{PSIC}_{n,(1,1)}$, which tests the equality of elements in U, is equivalent to the equality function EQ_n.

6.3 Index Function

We show that an index function, which takes a vector $D = (D_j)_{j\in[N]} \in \{0, 1\}^N$ and an index $x \in [N]$ as input and outputs D_x, admits ideal PSM. The index function was implicitly used to prove the feasibility of PSM for general functions f in [29] by setting D as the truth-table of $f(x_0, \cdot)$ and $x = x_1$. It has played an important role to improve the worst-case communication complexity of PSM [3, 7, 9, 39].

We here prove that a more general form of the index functions admit ideal PSM. Let H and G be abelian groups. Let $\mathcal{F}$ be a set of functions from H to G satisfying the following properties:

- For any $\phi, \phi' \in \mathcal{F}$, the function $\phi + \phi'$ (resp. $\phi - \phi'$), defined as $(\phi + \phi')(t) = \phi(t) + \phi'(t)$ (resp. $(\phi - \phi')(t) = \phi(t) - \phi'(t)$) for any $t \in H$, is also in $\mathcal{F}$.
- For any $\phi \in \mathcal{F}$ and $s \in H$, the function ϕ_s, defined as $\phi_s(t) = \phi(t - s)$ for any $t \in H$, is also in $\mathcal{F}$.

Set $X_0 = H \times G$ and $X_1 = \mathcal{F}$. Define $f : X_0 \times X_1 \to G$ as

$$f((x, r), \phi) = \phi(x) - r \tag{9}$$

for any $x \in H$, $r \in G$, and $\phi : H \to G \in \mathcal{F}$. The original index function can be viewed as a special instance by setting $G = \{0, 1\}$, $H = \mathbb{Z}_N$, and $\mathcal{F}$ as the set of all functions from $[N]$ to $\{0, 1\}$.

We show that f admits ideal PSM if it is non-degenerate. For $s \in H$, we define $(\pi^s_0, \pi^s_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ as

$$\pi^s_0(x, r) = (x + s, r),\ \pi^s_1(\phi) = \phi_s.$$

For $\tau \in X_1$, we define $(\sigma^\tau_0, \sigma^\tau_1)$ as

$$\sigma^\tau_0(x, r) = (x, r + \tau(x)),\ \sigma^\tau_1(\phi) = \phi + \tau.$$

Let S be a subgroup generated by $\{(\pi^s_0, \pi^s_1) : s \in H\} \cup \{(\sigma^\tau_0, \sigma^\tau_1) : \tau \in X_1\}$. From Theorem 4.5, it suffices to prove that the following conditions are satisfied.

(a) For any $(\pi_0, \pi_1) \in S$, $(x, r) \in X_0$ and $\phi \in X_1$, it holds that $f(\pi_0(x, r), \pi_1(\phi)) = f((x, r), \phi)$.

(b) For any $((x, r), \phi), ((x', r'), \phi')$ such that $f((x, r), \phi) = f((x', r'), \phi')$, there exists $(\pi_0, \pi_1) \in S$ such that $(\pi_0(x, r), \pi_1(\phi)) = ((x', r'), \phi')$.

The condition (a) is satisfied from the following equalities: For all $s \in H$ and $\tau \in X_1$,

$$f(\pi_0^s(x,r), \pi_1^s(\phi)) = \phi_s(x+s) - r = \phi(x) - r = f((x,r),\phi),$$
$$f(\sigma_0^\tau(x,r), \sigma_1^\tau(\phi)) = (\phi+\tau)(x) - r - \tau(x) = f((x,r),\phi).$$

Next, let $((x,r),\phi), ((x',r'),\phi') \in X_0 \times X_1$ be such that $f((x,r),\phi) = f((x',r'),\phi')$. We define $s \in H$ and $\tau \in X_1$ as

$$s = x' - x, \ \tau = \phi' - \phi_s.$$

We define $(\pi_0, \pi_1) \in S$ as $(\pi_0,\pi_1) = (\sigma_0^\tau, \sigma_1^\tau) \circ (\pi_0^s, \pi_1^s)$. Then, since $\phi'(x') - r' = \phi(x) - r$, we have that

$$\sigma_0^\tau \circ \pi_0^s(x,r) = \sigma_0^\tau(x+s, r) = (x+s, r+\tau(x+s)) = (x', r+\phi'(x') - \phi(x)) = (x', r'),$$
$$\sigma_1^\tau \circ \pi_1^s(\phi) = \sigma_1^\tau(\phi_s) = \phi_s + \tau.$$

Furthermore, $(\phi_s + \tau)(t) = \phi(t-s) + (\phi'(t) - \phi(t-s)) = \phi'(t)$ for any $t \in H$, and hence $\phi_s + \tau = \phi'$. Thus, the condition (b) is satisfied.

6.4 Inner Product

While the inner product itself does not admit ideal PSM, its variant that extends the domain admits ideal PSM. Formally, define $\mathsf{IP}_{q,r} : (\mathbb{F}_q^r \times \mathbb{F}_q) \times (\mathbb{F}_q^r \times \mathbb{F}_q) \to \mathbb{F}_q$ as

$$\mathsf{IP}_{q,r}((\mathbf{x}_0, a_0), (\mathbf{x}_1, a_1)) = \langle \mathbf{x}_0, \mathbf{x}_1 \rangle - a_0 - a_1$$

for any $\mathbf{x}_0, \mathbf{x}_1 \in \mathbb{F}_q^r$ and $a_0, a_1 \in \mathbb{F}_q$.

We show that $\mathsf{IP}_{q,r}$ is non-degenerate and admits ideal PSM. First, we prove that $\mathsf{IP}_{q,r}$ is non-degenerate. Assume for contradiction that $\mathsf{IP}_{q,r}$ is degenerate. Since $\mathsf{IP}_{q,r}$ is symmetric, we may assume that there exist $(\mathbf{x}_0, a_0) \neq (\mathbf{x}_0', a_0') \in \mathbb{F}_q^r \times \mathbb{F}_q$ such that $\mathsf{IP}_{q,r}((\mathbf{x}_0, a_0), (\mathbf{x}_1, a_1)) = \mathsf{IP}_{q,r}((\mathbf{x}_0', a_0'), (\mathbf{x}_1, a_1))$ for any $(\mathbf{x}_1, a_1) \in \mathbb{F}_q^r \times \mathbb{F}_q$. This implies that $\langle \mathbf{x}_0 - \mathbf{x}_0', \mathbf{x}_1 \rangle = a_0 - a_0'$. By setting $\mathbf{x}_1 = \mathbf{0}$, we have $a_0 = a_0'$. Since the inner product is non-degenerate, the fact that $\langle \mathbf{x}_0 - \mathbf{x}_0', \mathbf{x}_1 \rangle = 0$ for all $\mathbf{x}_1$ implies that $\mathbf{x}_0 = \mathbf{x}_0'$. This contradicts $(\mathbf{x}_0, a_0) \neq (\mathbf{x}_0', a_0')$.

To see that $\mathsf{IP}_{q,r}$ admits ideal PSM, we show that it can be expressed as a special instance of the index function f in Eq. (9). Let $G = \mathbb{F}_q$ and $H = \mathbb{F}_q^r$. For $\mathbf{x}_1 \in \mathbb{F}_q^r$ and $a_1 \in \mathbb{F}_q$, define a function $\phi^{\mathbf{x}_1, a_1} : H \to G$ as $\phi^{\mathbf{x}_1, a_1}(\mathbf{x}_0) = \langle \mathbf{x}_0, \mathbf{x}_1 \rangle - a_1$ for any $\mathbf{x}_0 \in H$. Let $X_0 = H \times G$ and $X_1 = \mathcal{F} = \{\phi^{\mathbf{x}_1, a_1} : \mathbf{x}_1 \in \mathbb{F}_q^r, a_1 \in \mathbb{F}_q\}$. It then holds that

$$\mathsf{IP}_{q,r}((\mathbf{x}_0, a_0), (\mathbf{x}_1, a_1)) = f((\mathbf{x}_0, a_0), \phi^{\mathbf{x}_1, a_1}).$$

We have that $\phi^{\mathbf{x}_1, a_1} + \phi^{\mathbf{x}_1', a_1'} = \phi^{\mathbf{x}_1 + \mathbf{x}_1', a_1 + a_1'}$ and $\phi_{\mathbf{s}}^{\mathbf{x}_1, a_1}(\mathbf{x}_0) := \phi^{\mathbf{x}_1, a_1}(\mathbf{x}_0 - \mathbf{s}) = \phi^{\mathbf{x}_1, a_1 + \langle \mathbf{s}, \mathbf{x}_1 \rangle}(\mathbf{x}_0)$. Hence, X_1 satisfies the assumption in Section 6.3. We thus conclude that $\mathsf{IP}_{q,r}$ admits ideal PSM.

6.5 Extension via Self-complementary Functions

Finally, we present an interesting method to transform a certain family of ideal PSM functions to a new family with extended input domains.

We prepare some notations. Let $f : [M_0] \times [M_1] \to \{0,1\}$ and $g : [N_0] \times [N_1] \to \{0,1\}$ be functions. We define a new function $h : ([M_0] \times [N_0]) \times ([M_1] \times [N_1]) \to \{0,1\}$ as

$$h((i,i'),(j,j')) = f(i,j) \oplus g(i',j')$$

for all $(i,i') \in [M_0] \times [N_0]$ and $(j,j') \in [M_1] \times [N_1]$. We denote this function by $f \boxplus g$. Intuitively, the truth-table of $f \boxplus g$ consists of $N_0 \times N_1$ blocks each of which is an M_0-by-M_1 matrix. For each $(i,j) \in [N_0] \times [N_1]$, the (i,j)-th block is equal to the truth-table of f if $g(i,j)=0$ and the truth-table of $\overline{f}$ if $g(i,j)=1$. For example, if f has the truth-table $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ and g has the truth-table $\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, then $f \boxplus g$ has the truth-table

$$\begin{pmatrix}1 & 0 & 0 & 1\\ 0 & 1 & 1 & 0\\ 0 & 1 & 1 & 0\\ 1 & 0 & 0 & 1\end{pmatrix}.$$

Note that $f \boxplus g \simeq g \boxplus f$.

We say that $f : [M_0] \times [M_1] \to \{0,1\}$ is *self-complementary* if there exist permutations $(\sigma_0, \sigma_1) \in \mathcal{S}_{M_0} \times \mathcal{S}_{M_1}$ such that

$$f(\sigma_0(i), \sigma_1(j)) = \overline{f}(i,j)$$

for all $(i,j) \in [M_0] \times [M_1]$. We define $\mathsf{rflip}_f \in \{0,1\}$ as $\mathsf{rflip}_f = 1$ if and only if there exist $i, i' \in [M_0]$ such that $f(i,j) \oplus f(i',j) = 1$ for all $j \in [M_1]$. That is, there exist two row vectors in the truth-table of f whose bit-wise XOR is the all-one vector. Similarly, we define $\mathsf{cflip}_f \in \{0,1\}$ as $\mathsf{cflip}_f = 1$ if and only if there exist $j, j' \in [M_1]$ such that $f(i,j) \oplus f(i,j') = 1$ for all $i \in [M_0]$. For $(a,b) \in \{0,1\}^2$, we say that f is (a,b)-*flippable* if $\mathsf{rflip}_f = a$ and $\mathsf{cflip}_f = b$. Note that if f is (a,b)-flippable, then the transpose $f^\top$ of f is (b,a)-flippable.

Now, we show the following theorem.

Theorem 6.1. *Let f be a non-degenerate, self-complementary, (a,b)-flippable function that admits ideal PSM and g be a non-degenerate, self-complementary, (a',b')-flippable function that admits ideal PSM. If $a \wedge a' = b \wedge b' = 0$, then $h = f \boxplus g$ is non-degenerate and admits ideal PSM. Furthermore, h is self-complementary and $(a \vee a', b \vee b')$-flippable.*

Proof. Let $X_0 \times X_1$ be a domain of f and let $Y_0 \times Y_1$ be a domain of g. First, we prove that $h\colon (X_0 \times Y_0) \times (X_1 \times Y_1) \to \{0,1\}$ is non-degenerate. Suppose to the contrary that h is degenerate. Suppose that there exist $(x_0, y_0) \neq (x_0', y_0') \in X_0 \times Y_0$ such that $h((x_0,y_0),(x_1,y_1)) = h((x_0',y_0'),(x_1,y_1))$ for all $(x_1,y_1) \in X_1 \times Y_1$. We have

$$f(x_0,x_1) \oplus g(y_0,y_1) = f(x_0',x_1) \oplus g(y_0',y_1) \;\Leftrightarrow\; f(x_0,x_1) \oplus f(x_0',x_1) = g(y_0,y_1) \oplus g(y_0',y_1)$$

for all $(x_1,y_1) \in X_1 \times Y_1$. Since the left-hand side of the second equality does not depend on y_1 and the right-hand side does not depend on x_0, there exists a constant $b \in \{0,1\}$ such that

$$f(x_0,x_1) \oplus f(x_0',x_1) = g(y_0,y_1) \oplus g(y_0',y_1) = b.$$

In the case of $b = 0$, this equality implies that either f or g is degenerate. Hence, we must have $b = 1$. In this case, we have $x_0 \neq x_0'$ and $y_0 \neq y_0'$ and therefore $\mathsf{rflip}_f = 1$ and $\mathsf{rflip}_g = 1$. However, this contradicts $a \wedge a' = 0$. Therefore, h is non-degenerate.

Next, we prove that h is self-complementary. Since f is self-complementary, there exist permutations $(\pi_0, \pi_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ such that

$$f(\pi_0(x_0), \pi_1(x_1)) = f(x_0,x_1) \oplus 1$$

for all $(x_0,x_1) \in X_0 \times X_1$. Define $\tau_0 \in \mathcal{S}_{X_0 \times Y_0}$ (resp. $\tau_1 \in \mathcal{S}_{X_0 \times Y_0}$) as $\tau_0(x_0,y_0) = (\pi_0(x_0), y_0)$ (resp. $\tau_1(x_1,y_1) = (\pi_1(x_1), y_1)$). We then have that

$$\begin{aligned} h(\tau_0(x_0,y_0), \tau_1(x_1,y_1)) &= h((\pi_0(x_0),y_0),(\pi_1(x_1),y_1)) \\ &= f(\pi_0(x_0),\pi_1(x_1)) \oplus g(y_0,y_1) \\ &= f(x_0,x_1) \oplus g(y_0,y_1) \oplus 1 \\ &= h((x_0,y_0),(x_1,y_1)) \oplus 1. \end{aligned}$$

Therefore, h is self-complementary.

Next, we prove that h is $(a \vee a', b \vee b')$-flippable. By symmetry, it suffices to prove that $\mathsf{rflip}_h = a \vee a' = \mathsf{rflip}_f \vee \mathsf{rflip}_g$. This is equivalent to proving that $\mathsf{rflip}_h = 1$ if and only if $\mathsf{rflip}_f = 1$ or $\mathsf{rflip}_g = 1$. If $\mathsf{rflip}_h = 1$, there exist $(x_0, y_0) \neq (x_0', y_0') \in X_0 \times Y_0$ such that for all $(x_1, y_1) \in X_1 \times Y_1$,

$$\begin{aligned} h((x_0, y_0), (x_1, y_1)) \oplus h((x_0', y_0'), (x_1, y_1)) = 1 &\Leftrightarrow f(x_0, x_1) \oplus f(x_0', x_1) \oplus g(y_0, y_1) \oplus g(y_0', y_1) = 1 \\ &\Leftrightarrow f(x_0, x_1) \oplus f(x_0', x_1) = g(y_0, y_1) \oplus g(y_0', y_1) \oplus 1. \end{aligned}$$

Since the left-hand side of the last equality does not depend on y_1 and the right-hand side does not depend on x_1, there exists a $b \in \{0, 1\}$ such that

$$f(x_0, x_1) \oplus f(x_0', x_1) = b, \; g(y_0, y_1) \oplus g(y_0', y_1) = b \oplus 1$$

for all (x_1, y_1). This implies that rflip_f or rflip_g is equal to 1. Hence, we have that $\mathsf{rflip}_h = 1$ implies that $\mathsf{rflip}_f = 1$ or $\mathsf{rflip}_g = 1$. On the other hand, if $\mathsf{rflip}_f = 1$, there exist $x_0 \neq x_0'$ such that for all $x_1 \in X_1$,

$$f(x_0, x_1) \oplus f(x_0', x_1) = 1.$$

Hence, we have

$$h((x_0, y_0), (x_1, y_1)) \oplus h((x_0', y_0), (x_1, y_1)) = 1$$

for all $y_0 \in Y_0$ and $(x_1, y_1) \in X_1 \times Y_1$, and this implies $\mathsf{rflip}_h = 1$. By the same argument, $\mathsf{rflip}_g = 1$ implies that $\mathsf{rflip}_h = 1$.

Finally, we prove that h admits ideal PSM. Since f and g are self-complementary, there exist $(\pi_0, \pi_1) \in \mathcal{S}_{X_0} \times \mathcal{S}_{Y_0}$ and $(\sigma_0, \sigma_1) \in \mathcal{S}_{X_1} \times \mathcal{S}_{Y_1}$ such that

$$f(\pi_0(x_0), \pi_1(x_1)) = f(x_0, x_1) \oplus 1, \; g(\sigma_0(y_0), \sigma_1(y_1)) = g(y_0, y_1) \oplus 1$$

for all $(x_0, x_1) \in X_0 \times X_1$ and $(y_0, y_1) \in Y_0 \times Y_1$. Since f and g admit ideal PSM, there exist subsets $S_f \subseteq \mathcal{S}_{X_0} \times \mathcal{S}_{X_1}$ and $S_g \subseteq \mathcal{S}_{Y_0} \times \mathcal{S}_{Y_1}$ satisfying the fourth condition of Theorem 4.5 for f and g, respectively. Define $S_h \subseteq \mathcal{S}_{X_0 \times Y_0} \times \mathcal{S}_{X_1 \times Y_1}$ as follows:

$$\begin{aligned} S_h = &\{(\phi_0^{\rho,\tau}, \phi_1^{\rho,\tau}) : \rho = (\rho_0, \rho_1) \in S_f, \tau = (\tau_0, \tau_1) \in S_g\} \\ &\cup \{(\psi_0^{\rho,\tau}, \psi_1^{\rho,\tau}) : \rho = (\rho_0, \rho_1) \in S_f, \tau = (\tau_0, \tau_1) \in S_g\}, \end{aligned}$$

where $\phi_i^{\rho,\tau} \in \mathcal{S}_{X_i \times Y_i}$ is a permutation that maps (x_i, y_i) to $(\rho_i(x_i), \tau_i(y_i))$ and $\psi_i^{\rho,\tau} \in \mathcal{S}_{X_i \times Y_i}$ is a permutation that maps (x_i, y_i) to $(\rho_i \circ \pi_i(x_i), \tau_i \circ \sigma_i(y_i))$. For $\rho = (\rho_0, \rho_1) \in S_f$, $\tau = (\tau_0, \tau_1) \in S_g$, and $((x_0, y_0), (x_1, y_1)) \in (X_0 \times Y_0) \times (X_1 \times Y_1)$, we have that

$$\begin{aligned} h(\phi_0^{\rho,\tau}((x_0, y_0)), \phi_1^{\rho,\tau}((x_1, y_1))) &= h((\rho_0(x_0), \tau_0(y_0), (\rho_1(x_1), \tau_1(y_1))) \\ &= f(\rho_0(x_0), \rho_1(x_1)) \oplus g(\tau_0(y_0), \tau_1(y_1)) \\ &= f(x_0, x_1) \oplus g(y_0, y_1) \\ &= h((x_0, y_0), (x_1, y_1)), \\ h(\psi_0^{\rho,\tau}((x_0, y_0)), \psi_1^{\rho,\tau}((x_1, y_1))) &= f(\rho_0 \circ \pi_0(x_0), \rho_1 \circ \pi_1(x_1)) \oplus g(\tau_0 \circ \sigma_0(y_0), \tau_1 \circ \sigma_1(y_1)) \\ &= f(\pi_0(x_0), \pi_1(x_1)) \oplus g(\sigma(y_0), \sigma(y_1)) \\ &= f(x_0, x_1) \oplus 1 \oplus g(y_0, y_1) \oplus 1 \\ &= h((x_0, y_0), (x_1, y_1)). \end{aligned}$$

On the other hand, let $((x_0, y_0), (x_1, y_1)), ((x_0', y_0'), (x_1', y_1')) \in (X_0 \times Y_0) \times (X_1 \times Y_1)$ be such that $h((x_0, y_0), (x_1, y_1)) = h((x_0', y_0'), (x_1', y_1'))$. Then, either of the following cases holds:

- $f(x_0, x_1) = f(x_0', x_1')$ and $g(y_0, y_1) = g(y_0', y_1')$; or

- $f(x_0, x_1) = f(x'_0, x'_1) \oplus 1$ and $g(y_0, y_1) = g(y'_0, y'_1) \oplus 1$.

In the first case, there exist $\rho = (\rho_0, \rho_1) \in S_f$ and $\tau = (\tau_0, \tau_1) \in S_g$ such that

$$\rho_0(x_0) = x'_0,\ \rho_1(x_1) = x'_1,\ \tau_0(y_0) = y'_0, \text{ and } \tau_1(y_1) = y'_1.$$

In the second case, there exist $\rho = (\rho_0, \rho_1) \in S_f$ and $\tau = (\tau_0, \tau_1) \in S_g$ such that

$$\rho_0 \circ \pi_0(x_0) = x'_0,\ \rho_1 \circ \pi_1(x_1) = x'_1,\ \tau_0 \circ \sigma_0(y_0) = y'_0, \text{ and } \tau_1 \circ \sigma_1(y_1) = y'_1.$$

This is because $f(\pi_0(x_0), \pi_1(x_1)) = f(x'_0, x'_1)$ and $g(\sigma_0(y_0), \sigma_1(y_1)) = g(y'_0, y'_1)$. Hence, in both cases, there exists a permutation in S_h that maps $((x_0, y_0), (x_1, y_1))$ to $((x'_0, y'_0), (x'_1, y'_1))$. Therefore, S_h satisfies the fourth condition of Theorem 4.5 for h. ☐

Let $\phi_{1,0} : [6] \times [10] \to \{0, 1\}$ be a function whose truth-table is

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

We show by a brute-force check that $\phi_{1,0}$ is self-complementary, $(0, 0)$-flippable, and an ideal PSM function. Let $\phi_{0,1} : [10] \times [6] \to \{0, 1\}$ be the transpose of $\phi_{1,0}$. Clearly, $\phi_{0,1}$ also is self-complementary, $(0, 0)$-flippable, and an ideal PSM function. For non-negative integers r, s, we define $\phi_{r,s}$ as

$$\phi_{r,s} = \underbrace{\phi_{1,0} \boxplus \cdots \boxplus \phi_{1,0}}_{r \text{ times}} \boxplus \underbrace{\phi_{0,1} \boxplus \cdots \boxplus \phi_{0,1}}_{s \text{ times}}.$$

Thus, it follows from Theorem 6.1 that $\phi_{r,s}$ is self-complementary, $(0, 0)$-flippable, and an ideal PSM function. The domain of $\phi_{r,s}$ is $[6^r 10^s] \times [6^s 10^r]$.

By augmenting certain functions with $\phi_{r,s}$ via the operation $\boxplus$, we obtain new infinite families of ideal PSM functions. Specifically, we have the following proposition.

Proposition 6.2. *Define* $\mathsf{id}_1 : \{0\} \times \{0, 1\} \to \{0, 1\}$ *as* $\mathsf{id}_1(i, j) = j$ *and* $\mathsf{id}_0 = \mathsf{id}_1^\top$. *Let* $r, s \geq 0$ *and* $k, k' \geq 2$. *The following functions are non-degenerate, self-complementary, and admit ideal PSM:*

- $(0, 0)$-*flippable ones:* $\phi_{r,s} : [6^r 10^s] \times [6^s 10^r] \to \{0, 1\}$.
- $(0, 1)$-*flippable ones:*
  - $\mathsf{id}_1 \boxplus \phi_{r,s} : [6^r 10^s] \times [2 \cdot 6^s 10^r] \to \{0, 1\}$.
  - $\mathsf{PSIC}_{2k,(1,k)} \boxplus \phi_{r,s} : [2k 6^r 10^s] \times [\binom{2k}{k} 6^s 10^r] \to \{0, 1\}$.
- $(1, 0)$-*flippable ones: the transposes of* $(0, 1)$-*flippable ones.*
- $(1, 1)$-*flippable ones:*
  - $\mathsf{EQ}_2 \boxplus \phi_{r,s} : [2 \cdot 6^r 10^s] \times [2 \cdot 6^s 10^r] \to \{0, 1\}$.
  - $\mathsf{id}_0 \boxplus \mathsf{PSIC}_{2k,(1,k)} \boxplus \phi_{r,s} : [4k 6^r 10^s] \times [\binom{2k}{k} 6^s 10^r] \to \{0, 1\}$.
  - $\mathsf{PSIC}_{2k,(1,k)}^\top \boxplus \mathsf{PSIC}_{2k',(1,k')} \boxplus \phi_{r,s} : [2k' \binom{2k}{k} 6^r 10^s] \times [2k \binom{2k'}{k'} 6^s 10^r] \to \{0, 1\}$.

Proof. Clearly, id_1 is self-complementary, $(0,1)$-flippable, and an ideal PSM function since its truth-table is $(0\ 1)$. Thus, from Theorem 6.1, $\mathsf{id}_1 \boxplus \phi_{r,s}$ is self-complementary, $(0,1)$-flippable, and an ideal PSM function.

We can see that $\mathsf{PSIC}_{2k,(1,k)} : [2k] \times [\binom{2k}{k}] \to \{0,1\}$ is self-complementary, $(0,1)$-flippable, and an ideal PSM function. Indeed, we have already seen that $\psi := \mathsf{PSIC}_{2k,(1,k)}$ admits ideal PSM in Section 6.2. For each $A \in \binom{[2k]}{k}$, let $\mathbf{c}_A \in \{0,1\}^{2k}$ be a column vector of the truth-table of ψ corresponding to A, that is, $\mathbf{c}_A[i] = 1$ if and only if $i \in A$. Then, the column vector of the truth-table of $\overline{\psi}$ corresponding to A is equal to the column vector $\mathbf{c}_{\overline{A}}$ of ψ corresponding to $\overline{A}$. Therefore, ψ is self-complementary. Furthermore, for any $i, i' \in [2k]$, there exists $A \in \binom{[2k]}{k}$ such that $\{i, j\} \subseteq A$ since $k \geq 2$. Then, $\mathbf{r}_i[A] = \mathbf{r}_{i'}[A] = 1$, where $\mathbf{r}_i$ is the row vector of the truth-table of ψ corresponding to i. Since $\mathbf{c}_A \oplus \mathbf{c}_{\overline{A}}$ is all-one vector, ψ is $(0,1)$-flippable. Thus, from Theorem 6.1, $\mathsf{PSIC}_{2k,(1,k)} \boxplus \phi_{r,s}$ is self-complementary, $(0,1)$-flippable, and an ideal PSM function.

Clearly, EQ_2 is self-complementary, $(1,1)$-flippable, and an ideal PSM function. Thus, $\mathsf{EQ}_2 \boxplus \phi_{r,s}$ does so due to Theorem 6.1. Generally, if f is $(0,1)$-flippable and g is $(1,0)$-flippable, then $h = f \boxplus g$ is $(1,1)$-flippable. Since id_0 is $(1,0)$-flippable, $\mathsf{id}_0 \boxplus \mathsf{PSIC}_{2k,(1,k)}$ is $(1,1)$-flippable. Furthermore, $\mathsf{PSIC}^{\top}_{2k,(1,k)} \boxplus \mathsf{PSIC}_{2k',(1,k')}$ is $(1,1)$-flippable. Therefore, $\mathsf{id}_0 \boxplus \mathsf{PSIC}_{2k,(1,k)} \boxplus \phi_{r,s}$ and $\mathsf{PSIC}^{\top}_{2k,(1,k)} \boxplus \mathsf{PSIC}_{2k',(1,k')} \boxplus \phi_{r,s}$ are self-complementary, $(1,1)$-flippable, and an ideal PSM function. ☐

7 Communication-efficient and Simplified PSM Schemes

In this section, we improve the communication complexity of state-of-the-art PSM schemes for several functions. Our constructions follow the same template: We embed a target function f into a larger function f' that admits an ideal PSM scheme, from which we naturally derive a PSM scheme for f. Since the ideal PSM scheme is communication-optimal, the key question is how much we can restrict the domain of f' while still containing f as a restriction. An additional benefit is that the resulting PSM schemes are possibly simpler than existing constructions, since ideal PSM schemes possess well-organized structures associated with permutation groups.

We say that a function $f : X_0 \times X_1 \to Y$ is *embedded* into a function $f' : X'_0 \times X'_1 \to Y$ if there are injective functions $\tau_i : X_i \to X'_i$ such that

$$f(x_0, x_1) = f'(\tau_0(x_0), \tau_1(x_1))$$

for all $(x_0, x_1) \in X_0 \times X_1$. If f is embedded into f', then a PSM scheme for f' naturally induces a PSM scheme for f. Let $\Pi' = (\mathsf{Gen}', \mathsf{Msg}', \mathsf{Eval}')$ be a PSM scheme for f'. We construct a PSM scheme $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ for f as follows: $\mathsf{Gen} = \mathsf{Gen}'$, $\mathsf{Msg}(i, x_i, r) = \mathsf{Msg}'(i, \tau_i(x_i), r)$, and $\mathsf{Eval} = \mathsf{Eval}'$. The correctness of Π follows since $\mathsf{Eval}((\mathsf{Msg}(i, x_i, r))_{i \in \{0,1\}}) = \mathsf{Eval}'((\mathsf{Msg}'(i, \tau_i(x_i), r))_{i \in \{0,1\}}) = f'(\tau_0(x_0), \tau_1(x_1)) = f(x_0, x_1)$. The privacy of Π follows since the distribution of $(\mathsf{Msg}'(i, \tau_i(x_i), r))_{i \in \{0,1\}}$ depends only on $f'(\tau_0(x_0), \tau_1(x_1)) = f(x_0, x_1)$.

We introduce a construction template based on ideal PSM schemes.

Embedding-to-ideal-PSM

Given a function $f : [N_0] \times [N_1] \to Y$, construct a PSM scheme Π for f as follows:

1. Find a non-degenerate function $f' : [N'_0] \times [N'_1] \to Y'$ such that
   - f' admits an ideal PSM scheme Π'.
   - f is embedded into f'.
2. Let Π be the PSM scheme induced by Π'.

The communication complexity of Π is the same as that of Π', i.e., $\log |N'_0| + \log |N'_1|$.

For example, the inner product $\langle\cdot,\cdot\rangle$ of r-dimensional vectors over $\mathbb{F}_q$ is embedded into the function $\mathsf{IP}_{q,r}$ via $\mathsf{IP}_{q,r}((\mathbf{x}_0,0),(\mathbf{x}_1,0)) = \langle\mathbf{x}_0,\mathbf{x}_1\rangle$. As a result, the ideal PSM scheme for $\mathsf{IP}_{q,r}$ induces a PSM scheme for the inner product whose communication complexity is $2(r+1)\log q$. This reproves the results by [2, 3].

Interestingly, to the best of our knowledge, all the existing PSM schemes achieving the state-of-the-art communication complexity in different computation models follow the above embedding-to-ideal-PSM template (see Appendix A). It is an interesting open question whether the best communication complexity among all PSM schemes for a given function can be achieved by some PSM scheme following the embedding-to-ideal-PSM template.

7.1 Functions with Sparse/Dense Truth-tables

First, we consider functions with sparse truth-tables. Specifically, let $f : [N_0] \times [N_1] \to \{0,1\}$ be a function such that each row of the truth-table T_f has at most d ones for a small $d \ll N_1$. Prior to this work, the only general construction in [7] can apply, resulting in communication complexity of $O(\max\{\sqrt{N_0},\sqrt{N_1}\})$.

We show an efficient PSM scheme for f with communication complexity $O(d\log(N_1+d))$, resulting in a strict improvement over [7] when $d = o(\sqrt{N_1}/\log N_1)$. The same communication complexity can be attained when the columns of the truth-table are sparse. These schemes also apply to functions such that the rows (or the columns) are dense, that is, they contain at least $n-d$ ones for a small d, by taking the NOT of sparse functions.

Proposition 7.1. *Let $f : [N_0] \times [N_1] \to \{0,1\}$ be a non-degenerate function such that for any $x_0 \in X_0$, the number of ones in the row $T_f[x_0,*]$ of the truth-table T_f is at most d. Then, there exists a PSM scheme for f such that $|M_0| = \binom{N_1+d}{d}$ and $|M_1| = N_1 + d$.*

Proof. We show that f is embedded into an ideal PSM function $f' : [N_0'] \times [N_1'] \to \{0,1\}$ with $N_0' = \binom{N_1+d}{d}$ and $N_1' = N_1 + d$. Then, an ideal PSM scheme for f' induces a PSM scheme Π for f with the stated communication complexity.

We define $f'' : [N_0] \times [N_1'] \to \{0,1\}$ by specifying its truth-table $T_{f''}$ as follows. For each $x_0 \in [N_0]$, define the row vector $T_{f''}[x_0,*]$ of length $N_1' = N_1 + d$ by appending $d - w_{x_0}$ ones and w_{x_0} zeros to the right of $T_f[x_0,*]$, where w_{x_0} denotes the number of ones in $T_f[x_0,*]$. Clearly, f is embedded into f'' and each row of $T_{f''}$ contains exact d ones.

We set $f' = \mathsf{PSIC}_{N_1',(d,1)}$. Note that f' has a domain $[N_0'] \times [N_1']$ with $N_0' = \binom{N_1+d}{d}$ and $N_1' = N_1 + d$. Then, f'' is embedded into f' since the rows of the truth-table of f' contains any row vector of weight d. Therefore, f is embedded into f'. □

7.2 Functions with Low-Rank Truth-tables

Narayanan et al. [43] showed PSM schemes for functions that have bounded tiling numbers. Specifically, a k-tiling of a function $f : X_0 \times X_1 \to Y$ is a partition of $X_0 \times X_1$ into k monochromatic rectangles, i.e., a tuple of rectangles $(R_i)_{i\in[k]}$ such that for every $i \in [k]$, there exists $y_i \in Y$ such that $f(x_0,x_1) = y_i$ for all $(x_0,x_1) \in R_i$. The tiling number of a function f is defined as the smallest number k such that f has a k-tiling. They showed a PSM scheme for f with communication complexity $O(k\log|Y|)$, where k is the tiling number of f.

Let q be the smallest prime power larger than $|Y|$ (we can choose q so that $q = O(|Y|)$). Then, the truth-table T_f of a function $f : X_0 \times X_1 \to Y$ can be viewed as a $|X_0|$-by-$|X_1|$ matrix over $\mathbb{F}_q$. In the following, we present a PSM scheme for f with communication complexity $O(r\log q) = O(r\log|Y|)$, where r is the rank of T_f over $\mathbb{F}_q$. As is well known, it holds that $k \geq r$ since T_f can be represented as a sum of k rank-1 matrices if f has a k-tiling (e.g. [32]). Our construction thus achieves a more refined communication complexity than [43].

Proposition 7.2. *Let $f : [N_0] \times [N_1] \to \mathbb{F}_q$ be a non-degenerate function whose truth-table $T_f \in \mathbb{F}_q^{N_0 \times N_1}$ has rank r as a matrix over $\mathbb{F}_q$. Then, there exists a PSM scheme for f such that $|M_0| = |M_1| = q^r$.*

Proof. We show that f is embedded into the inner-product function $\mathsf{IP}_{q,r} : \mathbb{F}_q^r \times \mathbb{F}_q^r \to \mathbb{F}_q$. Then, the proposition follows since $\mathsf{IP}_{q,r}$ admits ideal PSM as shown in Section 6.4.

Since T_f has rank r, there exist two matrices $\mathbf{M}_0 \in \mathbb{F}_q^{N_0 \times r}$ and $\mathbf{M}_1 \in \mathbb{F}_q^{N_1 \times r}$ such that $T_f = \mathbf{M}_0 \mathbf{M}_1^\top$ as matrices over $\mathbb{F}_q$. In particular, it holds that $T_f[x_0, x_1] = \langle \mathbf{M}_0[x_0], \mathbf{M}_1[x_1] \rangle$ for any $(x_0, x_1) \in [N_0] \times [N_1]$, where $\mathbf{M}_i[x]$ denotes the x-th row vector of $\mathbf{M}_i$.

For each $i \in \{0,1\}$, define $\tau_i : [N_i] \to \mathbb{F}_q^r$ by $\tau_i(x) = \mathbf{M}_i[x]$ for any $x \in [N_i]$. From the above, we have $f(x_0, x_1) = \mathsf{IP}_{q,r}(\tau_0(x_0), \tau_1(x_1))$ for any (x_0, x_1). Since f is non-degenerate, the rows of $\mathbf{M}_i$ are pairwise distinct and hence τ_i is injective. Therefore, f is embedded into $\mathsf{IP}_{q,r}$ via the τ_i's. ☐

7.3 Simplified PSM for General Functions

Beimel et al. [7] showed a PSM scheme for a general function $f : [N] \times [N] \to \{0,1\}$ with communication complexity $O(\sqrt{N})$. According to the description in [3], the more precise communication complexity is at least $12\sqrt{N}$.[4] In the following, we present a PSM scheme for f with communication complexity $8\sqrt{N} + 2$, improving the constant factor in the dominant term of the state-of-the-art construction. An additional benefit is that our construction is simpler and more direct: the message and evaluation functions are given in closed form, unlike the hierarchical designs of [3, 7], which rely on nested invocations of PSM schemes for smaller functions. A detailed comparison is provided in the last paragraph of this section.

Let $f : [N] \times [N] \to \{0,1\}$ be a non-degenerate function. We assume that N is a square number. Otherwise, we embed f into a function f' with domain $[N'] \times [N']$ for the smallest square number N' with $N' > N$. Let $n = \sqrt{N}$.

We define a multi-linear function $\hat{f} \colon (\{0,1\}^n)^4 \to \{0,1\}$ as

$$\hat{f}(\mathbf{a}_0, \mathbf{b}_0, \mathbf{a}_1, \mathbf{b}_1) = \sum_{i_0, j_0, i_1, j_1 \in [n]} \mathbf{a}_0[i_0] \cdot \mathbf{b}_0[j_0] \cdot \mathbf{a}_1[i_1] \cdot \mathbf{b}_1[j_1] \cdot f(i_0 n + j_0, i_1 n + j_1) \tag{10}$$

for any $\mathbf{a}_0, \mathbf{b}_0, \mathbf{a}_1, \mathbf{b}_1 \in \{0,1\}^n$, where we denote the i-th bit of a vector $\mathbf{v}$ by $\mathbf{v}[i]$. For $x \in [N]$, we let $\mathbf{u}_x, \mathbf{v}_x \in \{0,1\}^n$ denote one-hot vectors such that $\mathbf{u}_x \otimes \mathbf{v}_x = \mathbf{e}_x$, where $\mathbf{e}_x \in \{0,1\}^N$ denotes a one-hot vector whose x-th element is one. It then holds that

$$\hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{u}_y, \mathbf{v}_y) = f(x, y)$$

for all $x, y \in [N]$.

Let $X = Y = (\{0,1\}^n)^4 \times \{0,1\}$. We define $G_f \colon X \times Y \to \{0,1\}$ as

$$\begin{aligned} &G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) \\ &= \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{x}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{x}_1, \mathbf{j}_1 \rangle \oplus \langle \mathbf{y}_0, \mathbf{i}_0 \rangle \oplus \langle \mathbf{y}_1, \mathbf{i}_1 \rangle \oplus a \oplus b. \end{aligned} \tag{11}$$

We will prove the following proposition.

Proposition 7.3. *f is embedded into G_f and G_f admits ideal PSM.*

Since $\hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{u}_y, \mathbf{v}_y) = f(x, y)$, we have

$$G_f((\mathbf{u}_x, \mathbf{v}_x, \mathbf{0}, \mathbf{0}, 0), (\mathbf{u}_y, \mathbf{v}_y, \mathbf{0}, \mathbf{0}, 0)) = f(x, y) \tag{12}$$

and hence f is embedded into G_f.

We will prove that G_f admits ideal PSM. For $\mathbf{t}_0, \mathbf{t}_1 \in \{0,1\}^n$, we define $\pi^{\mathbf{t}_0, \mathbf{t}_1} = (\pi_0^{\mathbf{t}_0, \mathbf{t}_1}, \pi_1^{\mathbf{t}_0, \mathbf{t}_1}) \in \mathcal{S}_X \times \mathcal{S}_Y$ as follows:

$$\begin{aligned} &\pi_0^{\mathbf{t}_0, \mathbf{t}_1}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a) = (\mathbf{x}_0 \oplus \mathbf{t}_0, \mathbf{x}_1 \oplus \mathbf{t}_1, \mathbf{i}_0, \mathbf{i}_1, a), \\ &\pi_1^{\mathbf{t}_0, \mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b) = \big(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1), \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1), b \oplus \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle\big), \end{aligned}$$

[4] The original paper [7] mainly focused on asymptotic analysis and did not provide explicit constants.

where ϕ_0, ϕ_1 are defined as follows:

$$\phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) = \bigoplus_{k \in [n]} \hat{f}(\mathbf{e}_k, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1)\mathbf{e}_k,$$
$$\phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1) = \bigoplus_{k \in [n]} \hat{f}(\mathbf{t}_0, \mathbf{e}_k, \mathbf{y}_0, \mathbf{y}_1)\mathbf{e}_k.$$

Note that we have

$$\langle \mathbf{x}_1, \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1)\rangle = \hat{f}(\mathbf{t}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1),\ \langle \mathbf{x}_0, \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1)\rangle = \hat{f}(\mathbf{x}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1).$$

For $\mathbf{s}_0, \mathbf{s}_1 \in \{0,1\}^n$, we define $\sigma^{\mathbf{s}_0,\mathbf{s}_1} = (\sigma_0^{\mathbf{s}_0,\mathbf{s}_1}, \sigma_1^{\mathbf{s}_0,\mathbf{s}_1}) \in \mathcal{S}_X \times \mathcal{S}_Y$ as follows:

$$\sigma_0^{\mathbf{s}_0,\mathbf{s}_1}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a) = \big(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0 \oplus \psi_0^{\mathbf{s}_1}(\mathbf{x}_0, \mathbf{x}_1), \mathbf{i}_1 \oplus \psi_1^{\mathbf{s}_0}(\mathbf{x}_0, \mathbf{x}_1), a \oplus \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{i}_0, \mathbf{s}_0\rangle \oplus \langle \mathbf{i}_1, \mathbf{s}_1\rangle\big),$$
$$\sigma_1^{\mathbf{s}_0,\mathbf{s}_1}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b) = (\mathbf{y}_0 \oplus \mathbf{s}_0, \mathbf{y}_1 \oplus \mathbf{s}_1, \mathbf{j}_0, \mathbf{j}_1, b),$$

where ψ_0, ψ_1 are defined as follows:

$$\psi_0^{\mathbf{s}_1}(\mathbf{x}_0, \mathbf{x}_1) = \bigoplus_{k \in [n]} \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{e}_k, \mathbf{s}_1)\mathbf{e}_k,$$
$$\psi_1^{\mathbf{s}_0}(\mathbf{x}_0, \mathbf{x}_1) = \bigoplus_{k \in [n]} \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{s}_0, \mathbf{e}_k)\mathbf{e}_k.$$

Note that we have

$$\langle \psi_0^{\mathbf{s}_1}(\mathbf{x}_0, \mathbf{x}_1), \mathbf{y}_0\rangle = \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{s}_1),\ \langle \psi_1^{\mathbf{s}_0}(\mathbf{x}_0, \mathbf{x}_1), \mathbf{y}_1\rangle = \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{s}_0, \mathbf{y}_1).$$

For $\mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1 \in \{0,1\}^n$ and $r \in \{0,1\}$, we define

$$\tau^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r} = (\tau_0^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}, \tau_0^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}) \in \mathcal{S}_X \times \mathcal{S}_Y$$

as follows:

$$\tau_0^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a) = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0 \oplus \mathbf{p}_0, \mathbf{i}_1 \oplus \mathbf{p}_1, a \oplus \langle \mathbf{x}_0, \mathbf{q}_0\rangle \oplus \langle \mathbf{x}_1, \mathbf{q}_1\rangle \oplus r),$$
$$\tau_1^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b) = (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0 \oplus \mathbf{q}_0, \mathbf{j}_1 \oplus \mathbf{q}_1, b \oplus \langle \mathbf{p}_0, \mathbf{y}_0\rangle \oplus \langle \mathbf{p}_1, \mathbf{y}_1\rangle \oplus r)$$

We simply denote

$$\chi^\rho = \tau^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r} \circ \sigma^{\mathbf{s}_0,\mathbf{s}_1} \circ \pi^{\mathbf{t}_0,\mathbf{t}_1}$$

for $\rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r)$. We define $S \subseteq \mathcal{S}_X \times \mathcal{S}_Y$ as

$$S = \left\{\chi^\rho : \rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r) \in (\{0,1\}^n)^8 \times \{0,1\}\right\}.$$

Lemma 7.4. *If $\chi^\rho = \chi^{\rho'}$, then $\rho = \rho'$.*

Proof. Let $\rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r)$ and $\rho' = (\mathbf{t}'_0, \mathbf{t}'_1, \mathbf{s}'_0, \mathbf{s}'_1, \mathbf{p}'_0, \mathbf{p}'_1, \mathbf{q}'_0, \mathbf{q}'_1, r')$. If $\chi^\rho = \chi^{\rho'}$, then we have that

$$\begin{aligned}
\chi^\rho(\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0) &= \big((\mathbf{t}_0, \mathbf{t}_1, \psi_0^{\mathbf{s}_1}(\mathbf{t}_0, \mathbf{t}_1) \oplus \mathbf{p}_0, \psi_1^{\mathbf{s}_0}(\mathbf{t}_0, \mathbf{t}_1) \oplus \mathbf{p}_1, \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{t}_0, \mathbf{q}_0\rangle \oplus \langle \mathbf{t}_1, \mathbf{q}_1\rangle \oplus r),\\
&\qquad (\mathbf{s}_0, \mathbf{s}_1, \phi_0^{\mathbf{t}_1}(\mathbf{0}, \mathbf{0}) \oplus \mathbf{q}_0, \phi_1^{\mathbf{t}_0}(\mathbf{0}, \mathbf{0}) \oplus \mathbf{q}_1, \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{0}, \mathbf{0}) \oplus \langle \mathbf{p}_0, \mathbf{s}_0\rangle \oplus \langle \mathbf{p}_1, \mathbf{s}_1\rangle \oplus r)\big)\\
&= \big((\mathbf{t}_0, \mathbf{t}_1, \psi_0^{\mathbf{s}_1}(\mathbf{t}_0, \mathbf{t}_1) \oplus \mathbf{p}_0, \psi_1^{\mathbf{s}_0}(\mathbf{t}_0, \mathbf{t}_1) \oplus \mathbf{p}_1, \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{t}_0, \mathbf{q}_0\rangle \oplus \langle \mathbf{t}_1, \mathbf{q}_1\rangle \oplus r),\\
&\qquad (\mathbf{s}_0, \mathbf{s}_1, \mathbf{q}_0, \mathbf{q}_1, \langle \mathbf{p}_0, \mathbf{s}_0\rangle \oplus \langle \mathbf{p}_1, \mathbf{s}_1\rangle \oplus r)\big).
\end{aligned}$$

Since $\chi^\rho(\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0) = \chi^{\rho'}(\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0)$, we obtain that $\mathbf{t}_i = \mathbf{t}'_i$, $\mathbf{s}_i = \mathbf{s}'_i$, and $\mathbf{q}_i = \mathbf{q}'_i$ for any $i \in \{0,1\}$. Hence, we also obtain that $\mathbf{p}_i = \mathbf{p}'_i$ for $i \in \{0,1\}$ and $r = r'$. Thus, $\rho = \rho'$. $\square$

Lemma 7.5. *It holds that $S \subseteq I_{G_f}$. That is, for any $\chi^\rho \in S$ and $(\overline{\mathbf{x}}, \overline{\mathbf{y}}) \in X \times Y$, it holds that $G_f(\chi^\rho(\overline{\mathbf{x}}, \overline{\mathbf{y}})) = G_f(\overline{\mathbf{x}}, \overline{\mathbf{y}})$.*

Proof. For any $\mathbf{t}_0, \mathbf{t}_1 \in \{0,1\}^n$, it holds that

$$\begin{aligned}
&G_f(\pi_0^{\mathbf{t}_0,\mathbf{t}_1}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), \pi_1^{\mathbf{t}_0,\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) \\
&= \hat{f}(\mathbf{x}_0 \oplus \mathbf{t}_0, \mathbf{x}_1 \oplus \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \\
&\quad \oplus \langle \mathbf{x}_0 \oplus \mathbf{t}_0, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \\
&\quad \oplus \langle \mathbf{x}_1 \oplus \mathbf{t}_1, \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1) \rangle \\
&\quad \oplus \langle \mathbf{i}_0, \mathbf{y}_0 \rangle \oplus \langle \mathbf{i}_1, \mathbf{y}_1 \rangle \oplus a \oplus b \\
&\quad \oplus \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1)) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle \\
&= \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \hat{f}(\mathbf{t}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \hat{f}(\mathbf{x}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \\
&\quad \oplus \langle \mathbf{x}_0, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \\
&\quad \oplus \langle \mathbf{x}_1, \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1) \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1) \rangle \\
&\quad \oplus \langle \mathbf{i}_0, \mathbf{y}_0 \rangle \oplus \langle \mathbf{i}_1, \mathbf{y}_1 \rangle \oplus a \oplus b \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle \\
&= G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) \oplus \hat{f}(\mathbf{t}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \hat{f}(\mathbf{x}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \\
&\quad \oplus \langle \mathbf{x}_0, \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \oplus \langle \mathbf{x}_1, \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \oplus \langle \mathbf{t}_0, \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1) \rangle \oplus \langle \mathbf{t}_1, \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1) \rangle \\
&= G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)),
\end{aligned}$$

Similarly, for any $\mathbf{s}_0, \mathbf{s}_1 \in \{0,1\}^n$, it holds that

$$G_f(\sigma_0^{\mathbf{s}_0,\mathbf{s}_1}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), \sigma_1^{\mathbf{s}_0,\mathbf{s}_1}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) = G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)).$$

Furthermore, for any $\mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1 \in \{0,1\}^n$ and $r \in \{0,1\}$, it holds that

$$\begin{aligned}
&G_f(\tau_0^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), \tau_1^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)) \\
&= \hat{f}(\mathbf{x}_0, \mathbf{x}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{x}_0, \mathbf{j}_0 \oplus \mathbf{q}_0 \rangle \oplus \langle \mathbf{x}_1, \mathbf{j}_1 \oplus \mathbf{q}_1 \rangle \oplus \langle \mathbf{i}_0 \oplus \mathbf{p}_0, \mathbf{y}_0 \rangle \oplus \langle \mathbf{i}_1 \oplus \mathbf{p}_1, \mathbf{y}_1 \rangle \\
&\quad \oplus a \oplus \langle \mathbf{x}_0, \mathbf{q}_0 \rangle \oplus \langle \mathbf{x}_1, \mathbf{q}_1 \rangle \oplus r \oplus b \oplus \langle \mathbf{p}_0, \mathbf{y}_0 \rangle \oplus \langle \mathbf{y}_1, \mathbf{p}_1 \rangle \oplus r \\
&= G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)).
\end{aligned}$$

Thus, we conclude that $\chi^\rho = \tau^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r} \circ \sigma^{\mathbf{s}_0,\mathbf{s}_1} \circ \pi^{\mathbf{t}_0,\mathbf{t}_1} \in I_{G_f}$. ☐

Lemma 7.6. *For any $(\overline{\mathbf{x}}, \overline{\mathbf{y}}) \in X \times Y$, it holds that $\mathrm{Orb}_S((\overline{\mathbf{x}}, \overline{\mathbf{y}})) = G_f^{-1}(G_f(\overline{\mathbf{x}}, \overline{\mathbf{y}}))$. That is, for any $(\overline{\mathbf{x}}, \overline{\mathbf{y}}), (\overline{\mathbf{x}}', \overline{\mathbf{y}}') \in X \times Y$ with $G_f(\overline{\mathbf{x}}, \overline{\mathbf{y}}) = G_f(\overline{\mathbf{x}}', \overline{\mathbf{y}}')$, there exists $\chi^\rho \in S$ such that $\chi^\rho(\overline{\mathbf{x}}, \overline{\mathbf{y}}) = (\overline{\mathbf{x}}', \overline{\mathbf{y}}')$. Furthermore, the permutation χ^ρ is uniquely determined by $(\overline{\mathbf{x}}, \overline{\mathbf{y}})$ and $(\overline{\mathbf{x}}', \overline{\mathbf{y}}')$.*

Proof. Let

$$\overline{\mathbf{x}} = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a),\ \overline{\mathbf{y}} = (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b),\ \overline{\mathbf{x}}' = (\mathbf{x}_0', \mathbf{x}_1', \mathbf{i}_0', \mathbf{i}_1', a'),\ \overline{\mathbf{y}}' = (\mathbf{y}_0', \mathbf{y}_1', \mathbf{j}_0', \mathbf{j}_1', b').$$

We define $\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1 \in \{0,1\}^n, r \in \{0,1\}$ as follows:

$$\begin{aligned}
\mathbf{t}_0 &= \mathbf{x}_0 \oplus \mathbf{x}'_0, \\
\mathbf{t}_1 &= \mathbf{x}_1 \oplus \mathbf{x}'_1, \\
\mathbf{s}_0 &= \mathbf{y}_0 \oplus \mathbf{y}'_0, \\
\mathbf{s}_1 &= \mathbf{y}_1 \oplus \mathbf{y}'_1, \\
\mathbf{p}_0 &= \mathbf{i}_0 \oplus \mathbf{i}'_0 \oplus \psi_0^{\mathbf{s}_1}(\mathbf{x}'_0, \mathbf{x}'_1), \\
\mathbf{p}_1 &= \mathbf{i}_1 \oplus \mathbf{i}'_1 \oplus \psi_1^{\mathbf{s}_0}(\mathbf{x}'_0, \mathbf{x}'_1), \\
\mathbf{q}_0 &= \mathbf{j}_0 \oplus \mathbf{j}'_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1), \\
\mathbf{q}_1 &= \mathbf{j}_1 \oplus \mathbf{j}'_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1), \\
r &= a \oplus a' \oplus \hat{f}(\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{i}_0, \mathbf{s}_0 \rangle \oplus \langle \mathbf{i}_1, \mathbf{s}_1 \rangle \oplus \langle \mathbf{x}'_0, \mathbf{q}_0 \rangle \oplus \langle \mathbf{x}'_1, \mathbf{q}_1 \rangle
\end{aligned}$$

Then, we have

$$\begin{aligned}
(\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a) &\xrightarrow{\pi_0^{\mathbf{t}_0,\mathbf{t}_1}} (\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{i}_0, \mathbf{i}_1, a) \\
&\xrightarrow{\sigma_0^{\mathbf{s}_0,\mathbf{s}_1}} \big(\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{i}_0 \oplus \psi_0^{\mathbf{s}_1}(\mathbf{x}'_0, \mathbf{x}'_1), \mathbf{i}_1 \oplus \psi_1^{\mathbf{s}_0}(\mathbf{x}'_0, \mathbf{x}'_1), \\
&\qquad a \oplus \hat{f}(\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{i}_0, \mathbf{s}_0 \rangle \oplus \langle \mathbf{i}_1, \mathbf{s}_1 \rangle\big) \\
&\xrightarrow{\tau_0^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}} (\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{i}'_0, \mathbf{i}'_1, a'),
\end{aligned}$$

and

$$\begin{aligned}
(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b) &\xrightarrow{\pi_1^{\mathbf{t}_0,\mathbf{t}_1}} \big(\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1), \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1), \\
&\qquad b \oplus \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle\big) \\
&\xrightarrow{\sigma_1^{\mathbf{s}_0,\mathbf{s}_1}} \big(\mathbf{y}'_0, \mathbf{y}'_1, \mathbf{j}_0 \oplus \phi_0^{\mathbf{t}_1}(\mathbf{y}_0, \mathbf{y}_1), \mathbf{j}_1 \oplus \phi_1^{\mathbf{t}_0}(\mathbf{y}_0, \mathbf{y}_1), \\
&\qquad b \oplus \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle\big) \\
&\xrightarrow{\tau_1^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r}} (\mathbf{y}'_0, \mathbf{y}'_1, \mathbf{j}'_0, \mathbf{j}'_1, b''),
\end{aligned}$$

where

$$b'' = b \oplus \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle \oplus \langle \mathbf{p}_0, \mathbf{y}'_0 \rangle \oplus \langle \mathbf{p}_1, \mathbf{y}'_1 \rangle \oplus r.$$

From Lemma 7.5, we have

$$G_f((\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{i}'_0, \mathbf{i}'_1, a'), (\mathbf{y}'_0, \mathbf{y}'_1, \mathbf{j}'_0, \mathbf{j}'_1, b'')) = G_f((\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a), (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b)),$$

and the right-hand side also equals to $G_f((\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{i}'_0, \mathbf{i}'_1, a'), (\mathbf{y}'_0, \mathbf{y}'_1, \mathbf{j}'_0, \mathbf{j}'_1, b'))$. Therefore, by the definition of G_f, we have $b'' = b'$. This implies that $\tau^{\mathbf{p}_0,\mathbf{p}_1,\mathbf{q}_0,\mathbf{q}_1,r} \circ \sigma^{\mathbf{s}_0,\mathbf{s}_1} \circ \pi^{\mathbf{t}_0,\mathbf{t}_1} \in S$ maps $(\overline{\mathbf{x}}, \overline{\mathbf{y}})$ to $(\overline{\mathbf{x}}', \overline{\mathbf{y}}')$. The last statement follows since the choice of $(\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r)$ is uniquely determined by $(\overline{\mathbf{x}}, \overline{\mathbf{y}})$ and $(\overline{\mathbf{x}}', \overline{\mathbf{y}}')$. □

Therefore, it follows from Lemmas 7.5 and 7.6, and Theorem 4.5 that G_f admits ideal PSM. This completes the proof of Proposition 7.3.

Furthermore, we show an explicit construction of an ideal PSM scheme Π for G_f in Figure 1. Observe that $(\mathsf{Msg}(0, \overline{\mathbf{x}}, \rho), \mathsf{Msg}(1, \overline{\mathbf{y}}, \rho)) = \chi^\rho(\overline{\mathbf{x}}, \overline{\mathbf{y}})$. Since $\chi^\rho \in I_{G_f}$ from Lemma 7.5, we have that $\mathsf{Eval}(\mathsf{Msg}(0, \overline{\mathbf{x}}, \rho), \mathsf{Msg}(1, \overline{\mathbf{y}}, \rho)) = G_f(\overline{\mathbf{x}}, \overline{\mathbf{y}})$. Thus, the correctness follows. To see privacy, let $(\overline{\mathbf{x}}, \overline{\mathbf{y}})$ be any input and $z = G_f(\overline{\mathbf{x}}, \overline{\mathbf{y}})$. We can see that the distribution of $(\mathsf{Msg}(0, \overline{\mathbf{x}}, \rho), \mathsf{Msg}(1, \overline{\mathbf{y}}, \rho))$ where $\rho \leftarrow \mathsf{Gen}()$ is the uniform distribution over the

PSM scheme Π

Notations.

- Let N be a square number and $n = \sqrt{N}$.
- Let $f\colon [N] \times [N] \to \{0,1\}$ be a non-degenerate function.
- Let $\hat{f}$ and G_f be functions defined in Eqs. (10) and (11), respectively.

$\mathsf{Gen}() \to \rho$**:** Output

$$\rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r),$$

where $\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1 \leftarrow_{\$} \{0,1\}^n$ and $r \leftarrow_{\$} \{0,1\}$.

$\mathsf{Msg}(0, \overline{\mathbf{x}}, \rho) \to \overline{\mathbf{x}}'$**:** Given

$$\overline{\mathbf{x}} = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{i}_0, \mathbf{i}_1, a) \text{ and } \rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r),$$

1. Compute $\mathbf{x}_0' = \mathbf{x}_0 \oplus \mathbf{t}_0$ and $\mathbf{x}_1' = \mathbf{x}_1 \oplus \mathbf{t}_1$.
2. Compute

$$\mathbf{i}_0' = \mathbf{i}_0 \oplus \mathbf{p}_0 \oplus \bigoplus_{k \in [n]} \hat{f}(\mathbf{x}_0', \mathbf{x}_1', \mathbf{e}_k, \mathbf{s}_1)\mathbf{e}_k,$$
$$\mathbf{i}_1' = \mathbf{i}_1 \oplus \mathbf{p}_1 \oplus \bigoplus_{k \in [n]} \hat{f}(\mathbf{x}_0', \mathbf{x}_1', \mathbf{s}_0, \mathbf{e}_k)\mathbf{e}_k.$$

3. Compute

$$a' = \hat{f}(\mathbf{x}_0', \mathbf{x}_1', \mathbf{s}_0, \mathbf{s}_1) \oplus \langle \mathbf{i}_0, \mathbf{s}_0 \rangle \oplus \langle \mathbf{i}_1, \mathbf{s}_1 \rangle \oplus \langle \mathbf{x}_0', \mathbf{q}_0 \rangle \oplus \langle \mathbf{x}_1', \mathbf{q}_1 \rangle \oplus r.$$

4. Output $\overline{\mathbf{x}}' = (\mathbf{x}_0', \mathbf{x}_1', \mathbf{i}_0', \mathbf{i}_1', a')$.

$\mathsf{Msg}(1, \overline{\mathbf{y}}, \rho) \to \overline{\mathbf{y}}'$**:** Given

$$\overline{\mathbf{y}} = (\mathbf{y}_0, \mathbf{y}_1, \mathbf{j}_0, \mathbf{j}_1, b) \text{ and } \rho = (\mathbf{t}_0, \mathbf{t}_1, \mathbf{s}_0, \mathbf{s}_1, \mathbf{p}_0, \mathbf{p}_1, \mathbf{q}_0, \mathbf{q}_1, r),$$

1. Compute $\mathbf{y}_0' = \mathbf{y}_0 \oplus \mathbf{s}_0$ and $\mathbf{y}_1' = \mathbf{y}_1 \oplus \mathbf{s}_1$.
2. Compute

$$\mathbf{j}_0' = \mathbf{j}_0 \oplus \mathbf{q}_0 \oplus \bigoplus_{k \in [n]} \hat{f}(\mathbf{e}_k, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1)\mathbf{e}_k,$$
$$\mathbf{j}_1' = \mathbf{j}_1 \oplus \mathbf{q}_1 \oplus \bigoplus_{k \in [n]} \hat{f}(\mathbf{t}_0, \mathbf{e}_k, \mathbf{y}_0, \mathbf{y}_1)\mathbf{e}_k.$$

3. Compute

$$b' = \hat{f}(\mathbf{t}_0, \mathbf{t}_1, \mathbf{y}_0, \mathbf{y}_1) \oplus \langle \mathbf{t}_0, \mathbf{j}_0 \rangle \oplus \langle \mathbf{t}_1, \mathbf{j}_1 \rangle \oplus \langle \mathbf{p}_0, \mathbf{y}_0' \rangle \oplus \langle \mathbf{p}_1, \mathbf{y}_1' \rangle \oplus r.$$

4. Output $\overline{\mathbf{y}}' = (\mathbf{y}_0', \mathbf{y}_1', \mathbf{j}_0', \mathbf{j}_1', b')$.

$\mathsf{Eval}(\overline{\mathbf{x}}', \overline{\mathbf{y}}') \to z$**:** Given $\overline{\mathbf{x}}'$ and $\overline{\mathbf{y}}'$, output $z = G_f(\overline{\mathbf{x}}', \overline{\mathbf{y}}')$.

Figure 1: An ideal PSM scheme for G_f

set $G_f^{-1}(z)$. Indeed, let $(\overline{\mathbf{u}}, \overline{\mathbf{v}}) \in G_f^{-1}(z)$ be any other input. From Lemma 7.6, there exists a unique permutation $\chi \in S$ such that $\chi(\overline{\mathbf{x}}, \overline{\mathbf{y}}) = (\overline{\mathbf{u}}, \overline{\mathbf{v}})$. Furthermore, it follows from Lemma 7.4 that the permutation χ uniquely determines ρ such that $\chi = \chi^\rho$. This implies that there exists unique randomness ρ such that $(\mathsf{Msg}(0, \overline{\mathbf{x}}, \rho), \mathsf{Msg}(1, \overline{\mathbf{y}}, \rho)) = (\overline{\mathbf{u}}, \overline{\mathbf{v}})$.

Since f is embedded into G_f via Eq. (12), the ideal PSM scheme Π in Figure 1 naturally induces a PSM scheme for f. In summary, we obtain the following theorem.

Theorem 7.7. *Let $f : [N] \times [N] \to \{0,1\}$ be a non-degenerate function. Then, the scheme described in Figure 1 is an ideal PSM scheme for G_f. In particular, it provides a PSM scheme for f with communication complexity $8\lceil\sqrt{N}\rceil + 2$.*

Detailed Comparison with [3, 7]. In [7], a PSM scheme for a general function $f : [N] \times [N] \to \{0,1\}$ with communication complexity $O(\sqrt{N})$ was presented. In essence, the PSM scheme proceeds as follows: Shared randomness consists of uniformly random vectors $\mathbf{r}_0, \mathbf{r}_1, \mathbf{s}_0, \mathbf{s}_1 \in \{0,1\}^n$, where $n = \sqrt{N}$. Then, a party with input x sends $\mathbf{u}_x \oplus \mathbf{r}_0$ and $\mathbf{v}_x \oplus \mathbf{r}_1$. The other party with input y sends $\mathbf{u}_y \oplus \mathbf{s}_0$ and $\mathbf{v}_y \oplus \mathbf{r}_1$. A key equality is

$$\begin{aligned}
&\hat{f}(\mathbf{u}_x \oplus \mathbf{r}_0, \mathbf{v}_x \oplus \mathbf{r}_1, \mathbf{u}_y \oplus \mathbf{s}_0, \mathbf{v}_y \oplus \mathbf{r}_1) \\
&= \hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{u}_y, \mathbf{s}_1) \oplus \hat{f}(\mathbf{u}_x, \mathbf{r}_1, \mathbf{u}_y, \mathbf{s}_1) \oplus \hat{f}(\mathbf{r}_0, \mathbf{v}_x, \mathbf{u}_y, \mathbf{s}_1) \oplus \hat{f}(\mathbf{r}_0, \mathbf{r}_1, \mathbf{u}_y, \mathbf{s}_1) \\
&\oplus \hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{s}_0, \mathbf{s}_1) \oplus \hat{f}(\mathbf{u}_x, \mathbf{r}_1, \mathbf{s}_0, \mathbf{s}_1) \oplus \hat{f}(\mathbf{r}_0, \mathbf{v}_x, \mathbf{s}_0, \mathbf{s}_1) \oplus \hat{f}(\mathbf{r}_0, \mathbf{r}_1, \mathbf{s}_0, \mathbf{s}_1) \\
&\oplus \hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{s}_0, \mathbf{v}_y) \oplus \hat{f}(\mathbf{u}_x, \mathbf{r}_1, \mathbf{s}_0, \mathbf{v}_y) \oplus \hat{f}(\mathbf{r}_0, \mathbf{v}_x, \mathbf{s}_0, \mathbf{v}_y) \oplus \hat{f}(\mathbf{r}_0, \mathbf{r}_1, \mathbf{s}_0, \mathbf{v}_y) \\
&\oplus \hat{f}(\mathbf{u}_x, \mathbf{r}_1, \mathbf{u}_y, \mathbf{v}_y) \oplus \hat{f}(\mathbf{r}_0, \mathbf{r}_1, \mathbf{u}_y, \mathbf{v}_y) \oplus \hat{f}(\mathbf{r}_0, \mathbf{v}_x, \mathbf{u}_y, \mathbf{v}_y) \oplus \hat{f}(\mathbf{u}_x, \mathbf{v}_x, \mathbf{u}_y, \mathbf{v}_y) .
\end{aligned}$$

An external party, who receives $\mathbf{u}_x \oplus \mathbf{r}_0$, $\mathbf{v}_x \oplus \mathbf{r}_1$, $\mathbf{u}_y \oplus \mathbf{s}_0$, and $\mathbf{v}_y \oplus \mathbf{r}_1$, can compute the left-hand side on his own. The last term in the right-hand side is what he wants to compute, namely $f(x,y)$. It thus suffices to reveal the sum of the other terms in the right-hand side to the external party. The sum of the terms in red can be computed by a PSM scheme for a function of the form $\langle \mathbf{w}_x, \mathbf{u}_y \rangle \oplus b$, where $\mathbf{w}_x$ is some n-dimensional vector determined by x and b is some random value. (Note that the addition by b is necessary to ensure that the external party learns the value of the total sum only.) In [3, Appendix B], a PSM scheme for the inner product of n-dimensional vectors was used to compute this function, which has communication complexity $2n+2$. Similarly, we can reduce the sum of the terms highlighted in yellow to a function of the form $\langle \mathbf{w}_x, \mathbf{v}_y \rangle \oplus b$, the sum of those highlighted in green to $\langle \mathbf{u}_x, \mathbf{w}_y \rangle \oplus b$, and the term highlighted in blue to $\langle \mathbf{v}_x, \mathbf{w}_y \rangle \oplus b$. Consequently, the total communication complexity is $4n + 4(2n+2) = 12n + 8$ bits. On the other hand, our PSM scheme achieves communication complexity $8n + 2$, improving the constant factor in the dominant term.

We observe that when computing the function $\langle D_x, \mathbf{u}_y \rangle \oplus b$, we can replace PSM for the inner product with the PSM scheme Π_{FKN} for the index function described in Figure 2, which has communication complexity $n + \log n + 1$. Applying this optimization to [3, 7] results in communication complexity $4n + 4(n + \log n + 1) = 8n + 4\log n + 4$. Nevertheless, our scheme is more communication-efficient as it further removes the additional factor of $O(\log n)$.

Additionally, the construction of [3, 7] is hierarchical since it internally invokes a PSM scheme for smaller functions (either the inner product or the index function). In contrast, our scheme in Figure 1 gives the message and evaluation functions in closed form, which makes the construction simpler and more direct.

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A Existing PSM Schemes Following the Embedding-to-ideal-PSM Construction

In this section, we demonstrate that the embedding-to-ideal-PSM construction is commonly adopted by existing PSM schemes.

Assume that $\Pi = (\mathsf{Gen}, \mathsf{Msg}, \mathsf{Eval})$ is a PSM scheme for a non-degenerate function $f : X_0 \times X_1 \to Y$. Then, f is embedded into the evaluation function $\mathsf{Eval} : M_0 \times M_1 \to Y$. Indeed, let r be an arbitrarily fixed randomness and let $X'_i \subseteq M_i$ be the image of X_i under $\mathsf{Msg}(i, \cdot, r)$, i.e., $X'_i = \mathsf{Msg}(i, X_i, r)$. Then, from the correctness of Π and Lemma 4.3, the restriction of Eval to $X'_0 \times X'_1$ is equivalent to f and hence f is embedded into Eval. We show that the state-of-the-art PSM schemes in various computation models have evaluation functions that admit ideal PSM, which means that these schemes follow the embedding-to-ideal-PSM construction.

A.1 PSM Schemes for Index Functions

Let $f : [N] \times [N] \to \{0, 1\}$ be a non-degenerate function. Feige et al. [29] showed a PSM scheme Π_{FKN} for f described in Figure 2.

The evaluation function Eval in Π_{FKN} is viewed as the index function described in Section 6.3 with $G = \{0, 1\}$, $H = \mathbb{Z}_N$, and $X_1 = \mathcal{F}$ being the set of all functions from H to G. Thus, Eval is an ideal PSM function. The communication complexity of Π_{FKN} is $\log |M_0| + \log |M_1| = N + \log N + 1$.

PSM scheme Π_{FKN}

$\mathsf{Gen}() \to r$**:**

1. Choose $(r_0, \ldots, r_{N-1}) \leftarrow_{\$} \{0,1\}^N$ and $s \leftarrow_{\$} [N]$.
2. Output $\rho = (r_0, \ldots, r_{N-1}, s)$.

$\mathsf{Msg}(0, x_0, \rho) \to m_0$**:** Given $x_0 \in [N]$ and $\rho = (r_0, \ldots, r_{N-1}, s)$,

1. Compute $x_0' = x_0 + s \bmod N$.
2. Output $m_0 := (x_0', r_{x_0'})$.

$\mathsf{Msg}(1, x_1, \rho) \to m_1$**:** Given $x_1 = (D_i)_{i \in [N]} \in \{0,1\}^N$ and $\rho = (r_0, \ldots, r_{N-1}, s)$,

1. Compute $v_i = D_{i'} \oplus r_i$ for all $i \in [N]$, where $i' = i - s \bmod N$.
2. Output $m_1 := (v_i)_{i \in [N]}$.

$\mathsf{Eval}(m_0, m_1) \to y$**:** Given $m_0 = (x, r)$ and $m_1 = (v_i)_{i \in [N]}$, output $v_x \oplus r$.

Figure 2: A PSM scheme Π_{FKN}

A.2 PSM Schemes for Polynomials

Let $n_0, \ldots, n_{k-1} \in \mathbb{N}$. Define $X_0 = \mathbb{F}_q^{n_0 \cdots n_{k-1}}$ and

$$X_1 = \{\mathbf{x}_0 \otimes \cdots \otimes \mathbf{x}_{k-1} : (\mathbf{x}_j)_{j \in [k]} \in \mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}}\}.$$

Define $f : X_0 \times X_1 \to \mathbb{F}_q$ as

$$f(\mathbf{p}, \mathbf{x}_0 \otimes \cdots \otimes \mathbf{x}_{k-1}) = \langle \mathbf{p}, \mathbf{x}_0 \otimes \cdots \otimes \mathbf{x}_{k-1} \rangle.$$

Then, f is non-degenerate since the inner product $\langle \cdot, \cdot \rangle$ is non-degenerate. Liu et al. [39] showed a PSM scheme Π_{LVW} for f described in Figure 3.

The evaluation function Eval in Π_{LVW} is viewed as the index function described in Section 6.3 with $G = \mathbb{F}_q$, $H = \mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}}$ and $X_1 = \mathcal{F}$ being the set of all multi-linear polynomials from $\mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}}$ to $\mathbb{F}_q$. Thus, Eval is an ideal PSM function. The communication complexity of Π_{LVW} is $\log|M_0| + \log|M_1| = (N + S)\log q$, where $N = \prod_{j \in [k]}(n_j + 1)$ and $S = \sum_{j \in [k]} n_j + 1$.

The scheme Π_{LVW} provides a PSM scheme for computing a degree-k polynomial $p(\mathbf{x}, \mathbf{y})$ since there is a vector $\mathbf{p}_{\mathbf{x}}$ determined by $\mathbf{x}$ such that $p(\mathbf{x}, \mathbf{y}) = \langle \mathbf{p}_{\mathbf{x}}, \underbrace{\mathbf{y} \otimes \cdots \otimes \mathbf{y}}_{k \text{ times}} \rangle$.

A.3 PSM Schemes for Arithmetic Formulas and Branching Programs

Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ be a non-degenerate function computed by an arithmetic formula of depth d over $\mathbb{F}_q$. Then, there exists a PSM scheme for f with the following features[5]:

- The scheme specifies a finite group G and a partition (S_0, S_1) of $[m]$, where $m = 2^{(1+\epsilon)d}$ and $\epsilon > 0$ is any constant.
- The message of party b is a tuple $(g_i^{(b)})_{i \in S_b}$, where $g_i^{(b)} \in G$.

[5] The domain of f is decomposed into $\mathbb{F}_q^{n_0} \times \mathbb{F}_q^{n_1}$ where $n = n_0 + n_1$, and $\mathbb{F}_q^{n_0}$ and $\mathbb{F}_q^{n_1}$ are the input spaces of the first and second party, respectively.

PSM scheme Π_{LVW}

Notations.

- Let $n_0, \ldots, n_{k-1} \in \mathbb{N}$.
- Let $X_0 = \mathbb{F}_q^{n_0 \cdots n_{k-1}}$ and $X_1 = \{\mathbf{x}_0 \otimes \cdots \otimes \mathbf{x}_{k-1} : (\mathbf{x}_j)_{j\in[k]} \in \mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}}\}$.
- Set $N = \prod_{j\in[k]}(n_j + 1)$.

$\mathsf{Gen}() \to r$**:**

1. Choose $\mathbf{b}_j \leftarrow_{\$} \mathbb{F}_q^{n_j}$ for each $j \in [k]$ and $\mathbf{g} \leftarrow_{\$} \mathbb{F}_q^N$.
2. Output $r := (\overline{\mathbf{b}} = (\mathbf{b}_j)_{j\in[k]}, \mathbf{g})$.

$\mathsf{Msg}(0, x_0, r) \to m_0$**:** Given $x_0 = \mathbf{p}$ and $r = (\overline{\mathbf{b}} = (\mathbf{b}_j)_{j\in[k]}, \mathbf{g})$,

1. Compute a multi-linear polynomial $h : \mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}} \to \mathbb{F}_q$ defined by
$$h((\mathbf{y}_j)_{j\in[k]}) := \langle \mathbf{p}, (\mathbf{y}_0 - \mathbf{b}_0) \otimes \cdots \otimes (\mathbf{y}_{k-1} - \mathbf{b}_{k-1})\rangle + \langle \mathbf{g}, (\mathbf{y}_0||1) \otimes \cdots \otimes (\mathbf{y}_{k-1}||1)\rangle,$$
where $\mathbf{y}_j||1$ denotes the vector obtained by padding 1 at the end of $\mathbf{y}_j$.
2. Output $m_0 := h$.

$\mathsf{Msg}(1, x_1, r) \to m_1$**:** Given $x_1 = \mathbf{x}_0 \otimes \cdots \otimes \mathbf{x}_{k-1}$ and $r = (\overline{\mathbf{b}} = (\mathbf{b}_j)_{j\in[k]}, \mathbf{g})$,

1. Compute $\mathbf{m}_1^1 = \overline{\mathbf{x}} + \overline{\mathbf{b}}$, where $\overline{\mathbf{x}} = (\mathbf{x}_j)_{j\in[k]} \in \mathbb{F}_q^{n_0} \times \cdots \times \mathbb{F}_q^{n_{k-1}}$, and
$$m_1^2 = \langle \mathbf{g}, ((\mathbf{x}_0 + \mathbf{b}_0)||1) \otimes \cdots \otimes ((\mathbf{x}_{k-1} + \mathbf{b}_{k-1})||1)\rangle.$$
2. Output $m_1 := (\mathbf{m}_1^1, m_1^2)$.

$\mathsf{Eval}(m_0, m_1) \to y$**:** Given $m_0 = h$ and $m_1 = (\mathbf{m}_1^1, m_1^2)$, output $h(\mathbf{m}_1^1) - m_1^2$.

Figure 3: A PSM scheme Π_{LVW}

- The evaluation function Eval computes the product $\prod_{i\in[m]} z_i$, where we set $z_i = g_i^{(0)}$ for $i \in S_0$ and $z_i = g_i^{(1)}$ for $i \in S_1$.

Since the scheme also specifies a bijection between the output of the product $\prod_{i\in[m]} z_i$ and the output of the arithmetic formula f, the function computed by the above PSM scheme is equivalent to f. Such a PSM scheme can be obtained by the known reduction from arithmetic formulas to the group product [19, 34]. Since the function Eval has the same form as the group product in Section 6.1, it admits ideal PSM. The communication complexity of the above scheme is $\log|M_0| + \log|M_1| = m \log|G| = O(2^{(1+\epsilon)d})$.

The PSM scheme for branching programs in [29] has an evaluation function Eval that is a composition of the group product with a function ϕ. Specifically, Eval is as follows:

- Let X be a group consisting of non-singular matrices such that (1) upper-left $(N-1) \times (N-1)$ submatrix is non-singular, (2) (N, N)-th entry is non-zero, and (3) (N, i)-th entry is 0 for $i \neq N$.[6]

[6] An integer N and a prime p is specified by the scheme. We omit the detailed description of these parameters because it is not necessary for the proof.

- We define $\mathsf{Mult} : X^{N+1} \times X^{N+1} \to X$ as

$$\mathsf{Mult}((z_i^{(0)})_{i\in[N+1]}, (z_i^{(1)})_{i\in[N+1]}) = \prod_{i=0}^{N}(z_i^{(0)} z_i^{(1)}),$$

and $\phi : X \to \{0,1\}$ as a function such that $\phi(z) = 0$ if the upper-right entry of z is 0 and $\phi(z) = 1$ otherwise. $\mathsf{Eval} : X^{N+1} \times X^{N+1} \to \{0,1\}$ is defined as $\mathsf{Eval} = \phi \circ \mathsf{Mult}$.

We define two subgroups $X', X'' \leq X$ as follows:

- $Q' \in X'$ if and only if (1) all diagonal entries of Q' are non-zero and (2) all non-diagonal entries of Q' other than (i,N)-th entry for $2 \leq i \leq N-1$ are 0.
- $Q'' \in X''$ if and only if (1) $(N-1) \times (N-1)$ upper-left submatrix of Q'' is non-singular, (2) the (N,N)-th entry is 1, and (3) the other entries are 0.

For $Q' \in X'$ and $Q'' \in X''$, we define a permutation $(\pi_{Q'}, \sigma_{Q''}) \in \mathcal{S}_{X^{N+1}} \times \mathcal{S}_{X^{N+1}}$ as

$$\pi_{Q'}(z_0, \dots, z_N) = (Q' z_0, z_1, \dots, z_N),$$
$$\sigma_{Q''}(z_0, \dots, z_N) = (z_0, \dots, z_{N-1}, z_N Q'').$$

Define S as a subgroup of $\mathcal{S}_{X^{N+1}} \times \mathcal{S}_{X^{N+1}}$ generated by $I_{\mathsf{Mult}} \cup \{(\pi_{Q'}, \sigma_{Q''}) : Q' \in X', Q'' \in X''\}$. We prove that S satisfies the fourth condition of Theorem 4.5 and hence Eval admits ideal PSM.

First, we prove that $S \subseteq I_{\mathsf{Eval}}$. It is sufficient to prove that generators of S belongs to I_{Eval}. Since $\mathsf{Eval} = \phi \circ \mathsf{Mult}$, I_{Mult} is a subset of I_{Eval}. From the second properties of Lemma 1 in [29], a permutation $(\pi_{Q'}, \sigma_{Q''})$ does not change the value of Eval, and therefore $\{(\pi_{Q'}, \sigma_{Q''}) : Q' \in X', Q'' \in X''\} \subseteq I_{\mathsf{Eval}}$.

Next, we prove that, for any $(x_0, x_1), (x'_0, x'_1) \in X^{N+1} \times X^{N+1}$ such that $\mathsf{Eval}(x_0, x_1) = \mathsf{Eval}(x'_0, x'_1)$ $(=: b)$, there exists a permutation $(\pi, \sigma) \in S$ such that $x'_0 = \pi(x_0)$ and $x'_1 = \sigma(x_1)$. Let $H \in X$ be a matrix such that $H = I + H_b$, where I is the identity matrix and H_b is a matrix such that the $(1,N)$-th entry is b and the rest of entries are 0. From Lemma 1 in [29], there exist $Q', R' \in X'$ and $Q'', R'' \in X''$ such that $Q'\mathsf{Mult}(x_0, x_1)Q'' = R'\mathsf{Mult}(x'_0, x'_1)R'' = H_b$. This implies that $\mathsf{Mult}(\pi_{Q'}(x_0), \sigma_{Q''}(x_1)) = \mathsf{Mult}(\pi_{R'}(x'_0), \sigma_{R''}(x'_1)) = H_b$. From the definition of I_{Mult} and the fact that Mult admits ideal PSM, there exists $(\pi', \sigma') \in I_{\mathsf{Mult}}$ such that $\pi'(\pi_{Q'}(x_0)) = \pi_{R'}(x'_0)$ and $\sigma'(\sigma_{Q'}(x_1)) = \sigma_{R'}(x'_1)$. Hence, we have $\pi_{R'}^{-1} \circ \pi' \circ \pi_{Q'}(x_0) = x'_0$ and $\sigma_{R'}^{-1} \circ \sigma' \circ \sigma_{Q'}(x_1) = x'_1$. Since $(\pi_{R'}^{-1} \circ \pi' \circ \pi_{Q'}, \sigma_{R'}^{-1} \circ \sigma' \circ \sigma_{Q'}) \in S$, this completes the proof.

Table 1: Numbers of non-degenerate, connected, ideal PSM functions $f : X_0 \times X_1 \to Y$ with $\lvert X_0 \rvert \leq \lvert X_1 \rvert \leq 15$ except for $\lvert X_0 \rvert = \lvert X_1 \rvert = 12$.

$\lvert X_0 \rvert$ \ $\lvert X_1 \rvert$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	**1**	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2		**1**	0	0	0	0	0	0	0	0	0	0	0	0	0
3			**2**	0	0	**1**	0	0	0	0	0	0	0	0	0
4				**5**	0	**1**	0	**1**	0	0	0	**2**	0	0	0
5					**3**	0	0	0	0	**3**	0	0	0	0	0
6						**9**	0	**1**	**2**	**1**	0	**7**	0	0	**2**
7							**5**	0	0	0	0	0	0	**2**	0
8								**29**	0	0	0	**6**	0	**1**	0
9									**14**	0	0	**2**	0	0	0
10										**15**	0	**1**	0	0	**3**
11											**5**	0	0	0	0
12												N/A	0	0	**1**
13													**7**	0	0
14														**21**	0
15															**28**

Table 2: Numbers of non-degenerate, connected, ideal PSM functions $f : X_0 \times X_1 \to \{0, 1\}$ with $|X_0| \leq |X_1| \leq 23$.

$\|X_0\|$ \ $\|X_1\|$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
1	**1**	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2		**1**	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3			**1**	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4				**2**	0	**1**	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5					**1**	0	0	0	0	**1**	0	0	0	0	0	0	0	0	0	0	0	0	0
6						**1**	0	**1**	**1**	**1**	0	0	0	0	**1**	0	0	0	0	**1**	0	0	0
7							**2**	0	0	0	0	0	0	0	0	0	0	0	0	0	**1**	0	0
8								**2**	0	0	0	**1**	0	**1**	0	**2**	0	0	0	0	0	0	0
9									**2**	0	0	**1**	0	0	0	0	0	0	0	0	0	0	0
10										**1**	0	**1**	0	0	**1**	**1**	0	0	0	0	0	0	0
11											**2**	0	0	0	0	0	0	0	0	0	0	0	0
12												**3**	0	0	0	**5**	0	0	0	**2**	0	**1**	0
13													**2**	0	0	0	0	0	0	0	0	0	0
14														**1**	0	0	0	0	0	0	0	0	0
15															**3**	0	0	**1**	0	0	0	0	0
16																**4**	0	0	0	**2**	0	0	0
17																	**1**	0	0	0	0	0	0
18																		**4**	0	0	0	0	0
19																			**1**	0	0	0	0
20																				**1**	0	0	0
21																					**2**	0	0
22																						**1**	0
23																							**1**